

CHAPTER

11

World Climate and Climate Change

The world climate can be studied by organising information and data on climate and synthesising them in smaller units for easy understanding, description and analysis. Three broad approaches have been adopted for classifying climate. They are empirical, genetic and applied. Empirical classification is based on observed data, particularly on temperature and precipitation. Genetic classification attempts to organise climates according to their causes. Applied classification is for specific purpose.

Koeppen's Scheme of Classification of Climate

The most widely used classification of climate is the empirical climate classification scheme developed by V. Koeppen. Koeppen identified a close relationship between the distribution of vegetation and climate. He selected certain values of temperature and precipitation and related them to the distribution of vegetation and used these values for classifying the climates. It is an empirical classification based on mean annual and mean monthly temperature and precipitation data. He introduced the use of capital and small letters to designate climatic groups and types. Although developed in 1918 and modified over a period of time, Koeppen's scheme is still popular and in use.

Koeppen recognised five major climatic groups, four of them are based on temperature and one on precipitation. Table 11.1 lists the climatic groups and their characteristics according to Koeppen. The capital letters : A,C, D and E delineate humid climates and B dry climates.

The climatic groups are subdivided into types, designated by small letters, based on seasonality of precipitation and temperature characteristics. The seasons of dryness are indicated by the small letters : f, m, w and s, where f corresponds to no dry season,

Table 11.1 : Climatic Groups According to Koeppen

Group	*Characteristics*
A - Tropical	Average temperature of the coldest month is 18 C or higher
B - Dry Climates	Potential evaporation exceeds precipitation
C - Warm Temperate	The average temperature of the coldest month of the (Mid-latitude) climates years is higher than minus 3 C but below 18 C
D - Cold Snow Forest Climates	The average temperature of the coldest month is minus 3 C or below
E - Cold Climates	Average temperature for all months is below 10 C

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m - monsoon climate, w- winter dry season and s - summer dry season. The small letters a, b, c and d refer to the degree of severity of temperature. The B- Dry Climates are subdivided using the capital letters S for steppe or semi-arid and W for deserts. The climatic types are listed in Table 11.2. The distribution of climatic groups and types is shown in Table 11.1.

islands of East Indies. Significant amount of rainfall occurs in every month of the year as thunder showers in the afternoon. The temperature is uniformly high and the annual range of temperature is negligible. The maximum temperature on any day is around 30 C while the minimum temperature is around 20 C. Tropical evergreen forests with dense canopy cover and large biodiversity are found in this climate.

Table 11.2 : Climatic Types According to Koeppen

Group	*Type*	*Letter Code*	*Characteristics*
A-Tropical Humid Climate	Tropical wet	Af	No dry season
	Tropical monsoon	Am	Monsoonal, short dry season
	Tropical wet and dry	Aw	Winter dry season
B-Dry Climate	Subtropical steppe	BSh	Low-latitude semi arid or dry
	Subtropical desert	BWh	Low-latitude arid or dry
	Mid-latitude steppe	BSk	Mid-latitude semi arid or dry
	Mid-latitude desert	BWk	Mid-latitude arid or dry
C-Warm temperate (Mid-latitude) Climates	Humid subtropical	Cfa	No dry season, warm summer
	Mediterranean	Cs	Dry hot summer
	Marine west coast	Cfb	No dry season, warm and cool summer
D-Cold Snow-forest Climates	Humid continental	Df	No dry season, severe winter
	Subarctic	Dw	Winter dry and very severe
E-Cold Climates	Tundra	ET	No true summer
	Polar ice cap	EF	Perennial ice

Group A : Tropical Humid Climates

Tropical humid climates exist between Tropic of Cancer and Tropic of Capricorn. The sun being overhead throughout the year and the presence of Inter Tropical Convergence Zone (ITCZ) make the climate hot and humid. Annual range of temperature is very low and annual rainfall is high. The tropical group is divided into three types, namely (i) Af- Tropical wet climate; (ii) Am - Tropical monsoon climate; (iii) Aw- Tropical wet and dry climate.

Tropical Wet Climate (Af)

Tropical wet climate is found near the equator. The major areas are the Amazon Basin in South America, western equatorial Africa and the

Tropical Monsoon Climate (Am)

Tropical monsoon climate (Am) is found over the Indian sub-continent, North Eastern part of South America and Northern Australia. Heavy rainfall occurs mostly in summer. Winter is dry. The detailed climatic account of this climatic type is given in the book on *India: Physical Environment*.

Tropical Wet and Dry Climate (Aw)

Tropical wet and dry climate occurs north and south of Af type climate regions. It borders with dry climate on the western part of the continent and Cf or Cw on the eastern part. Extensive Aw climate is found to the north and south of the Amazon forest in Brazil and adjoining parts

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of Bolivia and Paraguay in South America, Sudan and south of Central Africa. The annual rainfall in this climate is considerably less than that in Af and Am climate types and is variable also. The wet season is shorter and the dry season is longer with the drought being more severe. Temperature is high throughout the year and diurnal ranges of temperature are the greatest in the dry season. Deciduous forest and tree-shredded grasslands occur in this climate.

Dry Climates : B

Dry climates are characterised by very low rainfall that is not adequate for the growth of plants. These climates cover a very large area of the planet extending over large latitudes from 15 - 60 north and south of the equator. At low latitudes, from 15 - 30 , they occur in the area of subtropical high where subsidence and inversion of temperature do not produce rainfall. On the western margin of the continents, adjoining the cold current, particularly over the west coast of South America, they extend more equatorwards and occur on the coast land. In middle latitudes, from 35 - 60 north and south of equator, they are confined to the interior of continents where maritime-humid winds do not reach and to areas often surrounded by mountains.

Dry climates are divided into steppe or semi-arid climate (BS) and desert climate (BW). They are further subdivided as subtropical steppe (BSh) and subtropical desert (BWh) at latitudes from 15 - 35 and mid-latitude steppe (BSk) and mid-latitude desert (BWk) at latitudes between 35 - 60 .

Subtropical Steppe (BSh) and Subtropical Desert (BWh) Climates

Subtropical steppe (BSh) and subtropical desert (BWh) have common precipitation and temperature characteristics. Located in the transition zone between humid and dry climates, subtropical steppe receives slightly more rainfall than the desert, adequate enough for the growth of sparse grasslands. The rainfall in both the climates is highly variable. The variability in the rainfall affects the life in the steppe much more than in the desert, more often causing famine. Rain occurs in short intense thundershowers in deserts and is ineffective in building soil moisture. *Fog* is common in coastal deserts bordering cold currents. Maximum temperature in the summer is very high. The highest shade temperature of 58 C was recorded at *Al Aziziyah*, Libya on 13 September 1922. The annual and diurnal ranges of temperature are also high.

Warm Temperate (Mid-Latitude) Climates-C

Warm temperate (mid-latitude) climates extend from 30 - 50 of latitude mainly on the eastern and western margins of continents. These climates generally have warm summers with mild winters. They are grouped into four types: (i) Humid subtropical, i.e. dry in winter and hot in summer (Cwa); (ii) Mediterranean (Cs); (iii) Humid subtropical, i.e. no dry season and mild winter (Cfa); (iv) Marine west coast climate (Cfb).

Humid Subtropical Climate (Cwa)

Humid subtropical climate occurs poleward of Tropic of Cancer and Capricorn, mainly in North Indian plains and South China interior plains. The climate is similar to Aw climate except that the temperature in winter is warm.

Mediterranean Climate (Cs)

As the name suggests, Mediterranean climate occurs around Mediterranean sea, along the west coast of continents in subtropical latitudes between 30 - 40 latitudes e.g. — Central California, Central Chile, along the coast in south eastern and south western Australia. These areas come under the influence of sub tropical high in summer and westerly wind in winter. Hence, the climate is characterised by hot, dry summer and mild, rainy winter. Monthly average temperature in summer is around 25 C and in winter below 10 C. The annual precipitation ranges between 35 - 90 cm.

Humid Subtropical (Cfa) Climate

Humid subtropical climate lies on the eastern parts of the continent in subtropical latitudes. In this region the air masses are generally

unstable and cause rainfall throughout the year. They occur in eastern United States of America, southern and eastern China, southern Japan, northeastern Argentina, coastal south Africa and eastern coast of Australia. The annual averages of precipitation vary from 75-150 cm. Thunderstorms in summer and frontal precipitation in winter are common. Mean monthly temperature in summer is around 27 C, and in winter it varies from 5 -12 C. The daily range of temperature is small.

Marine West Coast Climate (Cfb)

Marine west coast climate is located poleward from the Mediterranean climate on the west coast of the continents. The main areas are: Northwestern Europe, west coast of North America, north of California, southern Chile, southeastern Australia and New Zealand. Due to marine influence, the temperature is moderate and in winter, it is warmer than for its latitude. The mean temperature in summer months ranges from 15 -20 C and in winter 4 -10 C. The annual and daily ranges of temperature are small. Precipitation occurs throughout the year. Precipitation varies greatly from 50-250cm.

Cold Snow Forest Climates (D)

Cold snow forest climates occur in the large continental area in the northern hemisphere between 40 -70 north latitudes in Europe, Asia and North America. Cold snow forest climates are divided into two types: (i) Df- cold climate with humid winter; (ii) Dw- cold climate with dry winter. The severity of winter is more pronounced in higher latitudes.

Cold Climate with Humid Winters (Df)

Cold climate with humid winter occurs poleward of marine west coast climate and mid latitude steppe. The winters are cold and snowy. The frost free season is short. The annual ranges of temperature are large. The weather changes are abrupt and short. Poleward, the winters are more severe.

Cold Climate with Dry Winters (Dw)

Cold climate with dry winter occurs mainly over Northeastern Asia. The development of pronounced winter anti cyclone and its weakening in summer sets in monsoon like reversal of wind in this region. Poleward summer temperatures are lower and winter temperatures are extremely low with many locations experiencing below freezing point temperatures for up to seven months in a year. Precipitation occurs in summer. The annual precipitation is low from 12-15 cm.

Polar Climates (E)

Polar climates exist poleward beyond 70 latitude. Polar climates consist of two types: (i) Tundra (ET); (ii) Ice Cap (EF).

Tundra Climate (ET)

The tundra climate (ET) is so called after the types of vegetation, like low growing mosses, lichens and flowering plants. This is the region of permafrost where the sub soil is permanently frozen. The short growing season and water logging support only low growing plants. During summer, the tundra regions have very long duration of day light.

Ice Cap Climate (EF)

The ice cap climate (EF) occurs over interior Greenland and Antartica. Even in summer, the temperature is below freezing point. This area receives very little precipitation. The snow and ice get accumulated and the mounting pressure causes the deformation of the ice sheets and they break. They move as icebergs that float in the Arctic and Antarctic waters. Plateau Station, Antarctica ,79 S, portray this climate.

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Climate Change

The earlier chapters on climate summarised our understanding of climate as it prevails now. The type of climate we experience now might be prevailing over the last 10,000 years with minor and occasionally wide fluctuations. The planet earth has witnessed many variations in climate since the beginning. Geological records show alteration of glacial and inter-glacial periods. The geomorphological features, especially in high altitudes and high latitudes, exhibit traces of advances and retreats of glaciers. The sediment deposits in glacial lakes also reveal the occurrence of warm and cold periods. The rings in the trees provide clues about wet and dry periods. Historical records describe the vagaries in climate. All these evidences indicate that change in climate is a natural and continuous process.

India also witnessed alternate wet and dry periods. Archaeological findings show that the Rajasthan desert experienced wet and cool climate around 8,000 B.C. The period 3,000-1,700 B.C. had higher rainfall. From about 2,000-1,700 B.C., this region was the centre of the Harappan civilisation. Dry conditions accentuated since then.

In the geological past, the earth was warm some 500-300 million years ago, through the Cambrian, Ordovician and Silurian periods. During the Pleistocene epoch, glacial and inter-glacial periods occurred, the last major peak glacial period was about 18,000 years ago. The present inter-glacial period started 10,000 years ago.

Climate in the recent past

Variability in climate occurs all the time. The nineties decade of the last century witnessed extreme weather events. The 1990s recorded the warmest temperature of the century and some of the worst floods around the world. The worst devastating drought in the Sahel region, south of the Sahara desert, from 1967-1977 is one such variability. During the 1930s, severe drought occurred in southwestern Great Plains of the United States, described as the *dust bowl*. Historical records of crop yield or crop failures, of floods and migration of people tell about the effects of changing climate. A number of times Europe witnessed warm, wet, cold and dry periods, the significant episodes were the warm and dry conditions in the tenth and eleventh centuries, when the Vikings settled in Greenland. Europe witnessed "Little Ice Age" from 1550 to about 1850. From about 1885-1940 world temperature showed an upward trend. After 1940, the rate of increase in temperature slowed down.

Causes of Climate Change

The causes for climate change are many. They can be grouped into astronomical and terrestrial causes. The astronomical causes are the changes in solar output associated with sunspot activities. Sunspots are dark and cooler patches on the sun which increase and decrease in a cyclical manner. According to some meteorologists, when the number of sunspots increase, cooler and wetter weather and greater storminess occur. A decrease in sunspot numbers is associated with warm and drier conditions. Yet, these findings are not statistically significant.

An another astronomical theory is Millankovitch oscillations, which infer cycles in the variations in the earth's orbital characteristics around the sun, the wobbling of the earth and the changes in the earth's axial tilt. All these alter the amount of insolation received from the sun, which in turn, might have a bearing on the climate.

Volcanism is considered as another cause for climate change. Volcanic eruption throws up lots of aerosols into the atmosphere. These aerosols remain in the atmosphere for a considerable period of time reducing the sun's radiation reaching the Earth's surface. After the recent Pinatoba and El Cion volcanic eruptions, the average temperature of the earth fell to some extent for some years.

The most important anthropogenic effect on the climate is the increasing trend in the concentration of greenhouse gases in the atmosphere which is likely to cause global warming.

Global Warming

Due to the presence of greenhouse gases, the atmosphere is behaving like a *greenhouse*. The atmosphere also transmits the incoming solar radiation but absorbs the vast majority of long wave radiation emitted upwards by the earth's surface. The gases that absorb long wave radiation are called greenhouse gases. The processes that warm the atmosphere are often collectively referred to as the *greenhouse effect*.

> The term *greenhouse* is derived from the analogy to a greenhouse used in cold areas for preserving heat. *A greenhouse* is made up of glass. The glass which is transparent to incoming short wave solar radiation is opaque to outgoing long wave radiation. The glass, therefore, allows in more radiation and prevents the long wave radiation going outside the glass house, causing the temperature inside the glasshouse structure warmer than outside. When you enter a car or a bus, during summers, where windows are closed, you feel more heat than outside. Likewise during winter the vehicles with closed doors and windows remain warmer than the temperature outside. This is another example of the greenhouse effect.

Greenhouse Gases(GHGs)

The primary GHGs of concern today are carbon dioxide (CO_2), Chlorofluorocarbons (CFCs), methane (CH_4), nitrous oxide (N_2O) and ozone (O_3). Some other gases such as nitric oxide (NO) and carbon monoxide (CO) easily react with GHGs and affect their concentration in the atmosphere.

The effectiveness of any given GHG molecule will depend on the magnitude of the increase in its concentration, its life time in the atmosphere and the wavelength of radiation that it absorbs. The chlorofluorocarbons (CFCs) are highly effective. *Ozone* which absorbs ultra violet radiation in the stratosphere is very effective in absorbing terrestrial radiation when it is present in the lower troposphere. Another important point to be noted is that the more time the GHG molecule remains in the atmosphere, the longer it will take for earth's atmospheric system to recover from any change brought about by the latter.

The *largest concentration* of GHGs in the atmosphere is *carbon dioxide*. The emission of CO_2 comes mainly from fossil fuel combustion (oil, gas and coal). Forests and oceans are the sinks for the carbon dioxide. Forests use CO_2 in their growth. So, deforestation due to changes in land use, also increases the concentration of Co_2. The time taken for atmospheric CO_2 to adjust to changes in sources to sinks is 20-50 years. It is rising at about 0.5 per cent annually. Doubling of concentration of CO_2 over pre-industrial level is used as an index for estimating the changes in climate in climatic models.

Chlorofluorocarbons (CFCs) are products of human activity. *Ozone* occurs in the stratosphere where ultra-violet rays convert oxygen into ozone. Thus, ultra violet rays do not reach the earth's surface. The CFCs which drift into the stratosphere destroy the ozone. Large depletion of ozone occurs over Antarctica. *The* depletion *of ozone concentration in the* stratosphere is called the *ozone hole*. This allows the ultra violet rays to pass through the troposphere.

International efforts have been initiated for reducing the emission of GHGs into the atmosphere. The most important one is the *Kyoto protocol* proclaimed in 1997. This protocol went into effect in 2005, ratified by 141 nations. Kyoto protocol bounds the 35 industrialised countries to reduce their emissions by the year 2012 to 5 per cent less than the levels prevalent in the year 1990.

The increasing trend in the concentration of GHGs in the atmosphere may, in the long run, warm up the earth. Once the global warming sets in, it will be difficult to reverse it. The effect of global warming may not be uniform everywhere. Nevertheless, the adverse effect due to global warming will adversely affect the life supporting system. Rise in the sea level due to melting of glaciers and ice-caps and thermal expansion of the sea may inundate large parts of the coastal area and islands, leading to social problems. This is another cause for serious concern for the world

community. Efforts have already been initiated to control the emission of GHGs and to arrest the trend towards global warming. Let us hope the world community responds to this challenge and adopts a lifestyle that leaves behind a livable world for the generations to come.

One of the major concerns of the world today is global warming. Let us look at how much the planet has warmed up from the temperature records.

The annual average near-surface air temperature of the world is approximately 14°C.

An increasing trend in temperature was discernible in the 20th century. The greatest warming of the 20th century was during the two periods, 1901-44 and 1977-99. Over each of these two periods, global temperatures rose by about 0.4°C. In between, there was a slight cooling, which was more marked in the Northern Hemisphere.

The globally averaged annual mean temperature at the end of the 20th century was about 0.6°C above that recorded at the end of the 19th century. The seven warmest years during the 1856-2000 were recorded in the last decade. The year 1998 was the warmest year, probably not only for the 20th century but also for the whole millennium.



Greenhouse gases rising alarmingly

Ancient Air Bubbles Buried In Antarctic Ice To Shed More Light On Global Warming

It has happened in the North Atlantic and may happen again. According to scientists, global warming could lead to prolonged chill

ICE AGE cometh

Air pollution biggest killer Southeast Asia, says WH

Write an explanatory note on "global warming".

Gangotri is shrinking 23m every year

Geneva: Himalayan glaciers, including the Gangotri, are receding at among the fastest rates in the world due to global warming, threatening water shortages for millions of people in India, China and Nepal, a leading conservation group said on Monday.

The Worldwide Fund for Nature (WWF) said in a new study that Himalayan glaciers were receding 10-15 metres per year on average and that the rate was accelerating as global warming increases.

In India, the Gangotri glacier is receding at an average rate of 23 metres per year, the study said.

"Himalayan glaciers are among the fastest retreating glaciers globally due to the effects of global warming," the WWF said in a statement. "This will eventually result in water shortages for hundreds of millions of people who rely on glacier-dependent rivers in India, China and Nepal," it said.

Himalayan glaciers feed seven of Asia's greatest rivers — Ganga, Indus, Brahmaputra, Salween, Mekong, Yangtze and Huang Ho

This image shows how the Gangotri glacier terminus has retracted since 1780. The contour lines are approximate. (Image by Jesse Allen, Earth Observatory; based on data provided by the ASTER Science Team)

Warming Arctic could affect global weather

This file photo shows dead fish lying on the dried bottom of the Ding An reservoir in China's Hainan island. An island on the edge of the vast Pacific, Hainan gets a large part of its rain during the typhoon season. The problem is, for two years now, there has not been a single typhoon, and

Table 12.1 : Components and Processes of the Water Cycle

Components	*Processes*
Water storage in oceans	Evaporation Evapotranspiration Sublimation
Water in the atmosphere	Condensation Precipitation
Water storage in ice and snow	Snowmelt runoff to streams
Surface runoff	Stream flow freshwater storage infiltration
Groundwater storage	Groundwater discharge springs

atmosphere, landsurface and subsurface and the organisms.

About 91 per cent of the planetary water is found in the oceans. The remaining is held as freshwater in glaciers and icecaps, groundwater sources, lakes, soil moisture, atmosphere, streams and within life. Nearly 59 per cent of the water that falls on land returns to the atmosphere through evaporation from over the oceans as well as from other places. The remainder runs-off on the surface, infiltrates into the ground or a part of it becomes glacier.

It is to be noted that the renewable water on the earth is constant while the demand is increasing tremendously. This leads to water crisis in different parts of the world — spatially and temporally. The pollution of river waters has further aggravated the crisis. How can you intervene in improving the water quality and augmenting the available quantity of water?

Relief of the Ocean Floor

The oceans are confined to the great depressions of the earth's outer layer. In this section, we shall see the nature of the ocean basins of the earth and their topography. The oceans, unlike the continents, merge so naturally into one another that it is hard to demarcate them. The geographers have divided the oceanic part of the earth into five oceans, namely the Pacific, the Atlantic, the Indian, Southern ocean and the Arctic. The various seas, bays, gulfs and other inlets are parts of these four large oceans.

A major portion of the ocean floor is found between 3-6 km below the sea level. The 'land' under the waters of the oceans, that is, the ocean floor exhibits complex and varied features as those observed over the land (Figure 12.2). The floors of the oceans are rugged with the world's largest mountain ranges, deepest trenches and the largest plains. These features are formed, like those of the continents, by the factors of tectonic, volcanic and depositional processes.

Divisions of the Ocean Floors

The ocean floors can be divided into four major divisions: (i) the Continental Shelf; (ii) the Continental Slope; (iii) the Deep Sea Plain; (iv) the Oceanic Deeps. Besides, these divisions there are also major and minor relief features in the ocean floors like ridges, hills, sea mounts, guyots, trenches, canyons, etc.

Continental Shelf

The continental shelf is the extended margin of each continent occupied by relatively shallow seas and gulfs. It is the shallowest part of the ocean showing an average gradient of 1° or even less. The shelf typically ends at a very steep slope, called the shelf break.

The width of the continental shelves vary from one ocean to another. The average width of continental shelves is about 80 km. The shelves are almost absent or very narrow along some of the margins like the coasts of Chile, the west coast of Sumatra, etc. On the contrary, the Siberian shelf in the Arctic Ocean, the largest in the world, stretches to 1,500 km in width. The depth of the shelves also varies. It may be as shallow as 30 m in some areas while in some areas it is as deep as 600 m.

The continental shelves are covered with variable thicknesses of sediments brought down by rivers, glaciers, wind, from the land and distributed by waves and currents. Massive sedimentary deposits received over a long time by the continental shelves, become the source of fossil fuels.

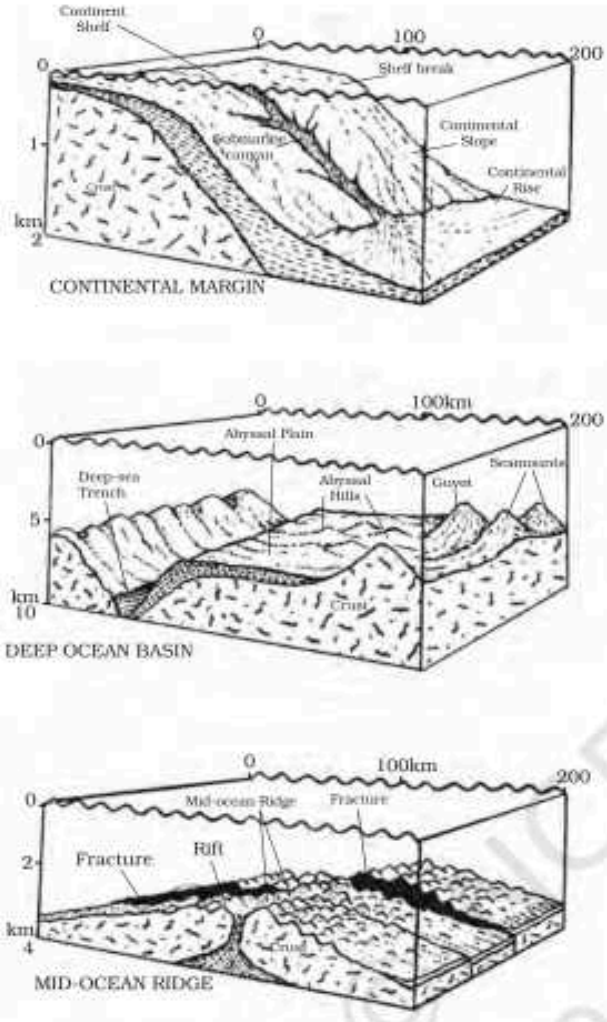


Figure 12.2 : Relief features of ocean floors

Continental Slope

The continental slope connects the continental shelf and the ocean basins. It begins where the bottom of the continental shelf sharply drops off into a steep slope. The gradient of the slope region varies between 2-5 . The depth of the slope region varies between 200 and 3,000 m. The slope boundary indicates the end of the continents. Canyons and trenches are observed in this region.

Deep Sea Plain

Deep sea plains are gently sloping areas of the ocean basins. These are the flattest and smoothest regions of the world. The depths vary between 3,000 and 6,000m. These plains are covered with fine-grained sediments like clay and silt.

Oceanic Deeps or Trenches

These areas are the deepest parts of the oceans. The trenches are relatively steep sided, narrow basins. They are some 3-5 km deeper than the surrounding ocean floor. They occur at the bases of continental slopes and along island arcs and are associated with active volcanoes and strong earthquakes. That is why they are very significant in the study of plate movements. As many as 57 deeps have been explored so far; of which 32 are in the Pacific Ocean; 19 in the Atlantic Ocean and 6 in the Indian Ocean.

Minor Relief Features

Apart from the above mentioned major relief features of the ocean floor, some minor but significant features predominate in different parts of the oceans.

Mid-Oceanic Ridges

A mid-oceanic ridge is composed of two chains of mountains separated by a large depression. The mountain ranges can have peaks as high as 2,500 m and some even reach above the ocean's surface. Iceland, a part of the mid-Atlantic Ridge, is an example.

Seamount

It is a mountain with pointed summits, rising from the seafloor that does not reach the surface of the ocean. Seamounts are volcanic in origin. These can be 3,000-4,500 m tall. The Emperor seamount, an extension of the Hawaiian Islands in the Pacific Ocean, is a good example.

Submarine Canyons

These are deep valleys, some comparable to the Grand Canyon of the Colorado river. They are sometimes found cutting across the continental shelves and slopes, often extending

from the mouths of large rivers. The Hudson Canyon is the best known submarine canyon in the world.

Guyots

It is a flat topped seamount. They show evidences of gradual subsidence through stages to become flat topped submerged mountains. It is estimated that more than 10,000 seamounts and guyots exist in the Pacific Ocean alone.

Atoll

These are low islands found in the tropical oceans consisting of coral reefs surrounding a central depression. It may be a part of the sea (lagoon), or sometimes form enclosing a body of fresh, brackish, or highly saline water.

Temperature of Ocean Waters

This section deals with the spatial and vertical variations of temperature in various oceans. Ocean waters get heated up by the solar energy just as land. The process of heating and cooling of the oceanic water is slower than land.

Factors Affecting Temperature Distribution

The factors which affect the distribution of temperature of ocean water are :

(i) *Latitude* : the temperature of surface water decreases from the equator towards the poles because the amount of insolation decreases poleward.

(ii) *Unequal distribution of land and water* : the oceans in the northern hemisphere receive more heat due to their contact with larger extent of land than the oceans in the southern hemisphere.

(iii) *Prevailing wind* : the winds blowing from the land towards the oceans drive warm surface water away form the coast resulting in the upwelling of cold water from below. It results into the longitudinal variation in the temperature. Contrary to this, the onshore winds pile up warm water near the coast and this raises the temperature.

(iv) *Ocean currents* : warm ocean currents raise the temperature in cold areas while the cold currents decrease the temperature in warm ocean areas. Gulf stream (warm current) raises the temperature near the eastern coast of North America and the West Coast of Europe while the Labrador current (cold current) lowers the temperature near the north-east coast of North America.

All these factors influence the temperature of the ocean currents locally. The enclosed seas in the low latitudes record relatively higher temperature than the open seas; whereas the enclosed seas in the high latitudes have lower temperature than the open seas.

Horizontal and Vertical Distribution of Temperature

The temperature-depth profile for the ocean water shows how the temperature decreases with the increasing depth. The profile shows a boundary region between the surface waters of the ocean and the deeper layers. The boundary usually begins around 100 - 400 m below the sea surface and extends several hundred of metres downward (Figure 12.3). This boundary region, from where there is a rapid decrease of temperature, is called the *thermocline*. About 90 per cent of the total volume of water is found below the thermocline in the deep ocean. In this zone, temperatures approach 0 C.

The temperature structure of oceans over middle and low latitudes can be described as a three-layer system from surface to the bottom.

The *first layer* represents the top layer of warm oceanic water and it is about 500m thick with temperatures ranging between 20 and 25 C. This layer, within the tropical region, is present throughout the year but in mid latitudes it develops only during summer.

The *second layer* called the thermocline layer lies below the first layer and is characterised by rapid decrease in temperature with increasing depth. The thermocline is 500 -1,000 m thick.

The *third layer* is very cold and extends upto the deep ocean floor. In the Arctic and

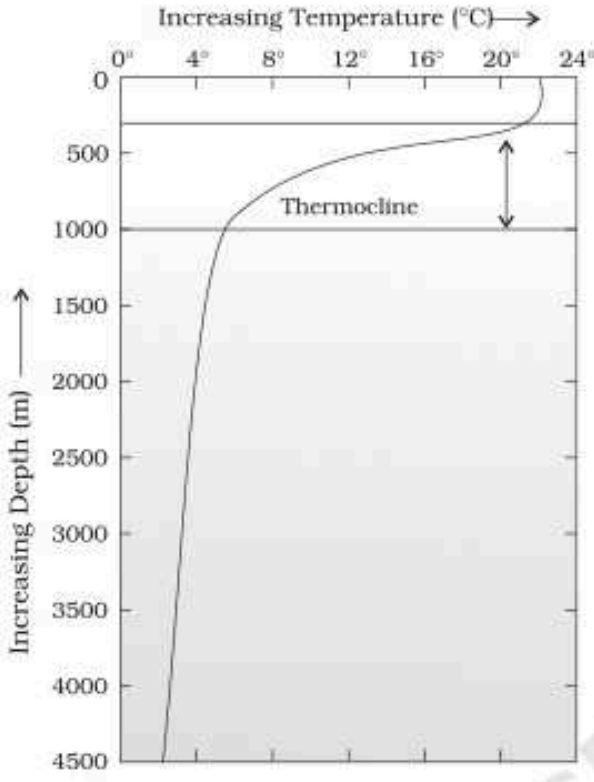


Figure 12.3 : Thermocline

Antartic circles, the surface water temperatures are close to 0 C and so the temperature change with the depth is very slight. Here, only one layer of cold water exists, which extends from surface to deep ocean floor.

The average temperature of surface water of the oceans is about 27 C and it gradually decreases from the equator towards the poles. The rate of decrease of temperature with increasing latitude is generally 0.5 C per latitude. The average temperature is around 22 C at 20 latitudes, 14 C at 40 latitudes and 0 C near poles. The oceans in the northern hemisphere record relatively higher temperature than in the southern hemisphere. The highest temperature is not recorded at the equator but slightly towards north of it. The average annual temperatures for the northern and southern hemisphere are around 19 C and 16 C respectively. This variation is due to the unequal distribution of land and water in the northern and southern hemispheres.

Figure 12.4 shows the spatial pattern of surface temperature of the oceans.

It is a well known fact that the maximum temperature of the oceans is always at their surfaces because they directly receive the heat from the sun and the heat is transmitted to the lower sections of the oceans through the process of convection. It results into decrease of temperature with the increasing depth, but the rate of decrease is not uniform throughout. The temperature falls very rapidly up to the depth of 200 m and thereafter, the rate of decrease of temperature is slowed down.

Salinity of Ocean Waters

All waters in nature, whether rain water or ocean water, contain dissolved mineral salts. Salinity is the term used to define the total content of dissolved salts in sea water (Table 12.4). It is calculated as the amount of salt (in gm) dissolved in 1,000 gm (1 kg) of seawater. It is usually expressed as parts per thousand ($^{o}/_{oo}$) or ppt. Salinity is an important property of sea water. Salinity of 24.7 $^{o}/_{oo}$ has been considered as the upper limit to demarcate 'brackish water'.

Factors affecting ocean salinity are mentioned below:

(i) The salinity of water in the surface layer of oceans depend mainly on evaporation and precipitation.

(ii) Surface salinity is greatly influenced in coastal regions by the fresh water flow from rivers, and in polar regions by the processes of freezing and thawing of ice.

(iii) Wind, also influences salinity of an area by transferring water to other areas.

(iv) The ocean currents contribute to the salinity variations. Salinity, temperature and density of water are interrelated. Hence, any change in the temperature or density influences the salinity of water in an area.

Highest salinity in water bodies
Lake Van in Turkey (330 $^{o}/_{oo}$),
Dead Sea (238 $^{o}/_{oo}$),
Great Salt Lake (220 $^{o}/_{oo}$)

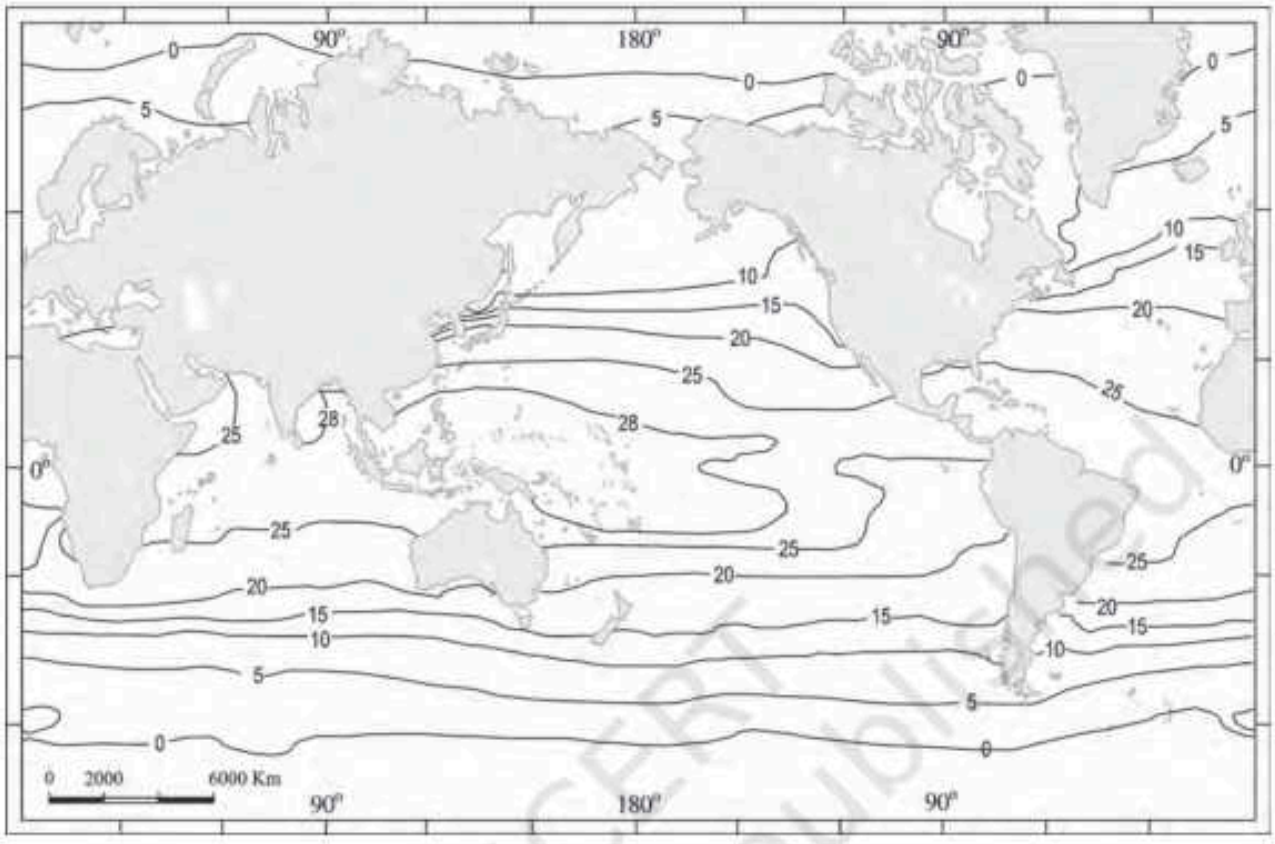


Figure 12.4 : Spatial pattern of surface temperature (C) of the oceans

Horizontal Distribution of Salinity

The salinity for normal open ocean ranges between 33°/$_{oo}$ and 37 °/$_{oo}$. In the land locked Red Sea, it is as high as 41°/$_{oo}$, while in the estuaries and the Arctic, the salinity fluctuates from 0 - 35 °/$_{oo}$, seasonally. In hot and dry regions, where evaporation is high, the salinity sometimes reaches to 70 °/$_{oo}$.

The salinity variation in the Pacific Ocean is mainly due to its shape and larger areal extent. Salinity decreases from 35 °/$_{oo}$ - 31 °/$_{oo}$ on the western parts of the northern hemisphere because of the influx of melted water from the Arctic region. In the same way, after 15 - 20 south, it decreases to 33 °/$_{oo}$.

The average salinity of the Atlantic Ocean is around 36 °/$_{oo}$. The highest salinity is recorded between 15 and 20 latitudes. Maximum salinity (37 °/$_{oo}$) is observed between 20 N and 30 N and 20 W - 60 W. It gradually decreases towards the north. The North Sea, in spite of its location in higher latitudes, records higher salinity due to more saline water brought by the North Atlantic Drift. Baltic Sea records low salinity due to influx of river waters in large quantity. The Mediterranean Sea records higher salinity due to high evaporation. Salinity is, however, very low in Black Sea due to enormous fresh water influx by rivers. See the atlas to find out the rivers joining Black Sea.

The average salinity of the Indian Ocean is 35 °/$_{oo}$. The low salinity trend is observed in the Bay of Bengal due to influx of river water. On the contrary, the Arabian Sea shows higher salinity due to high evaporation and low influx of fresh water. Figure 12.5 shows the salinity of the World's oceans.

Vertical Distribution of Salinity

Salinity changes with depth, but the way it changes depends upon the location of the

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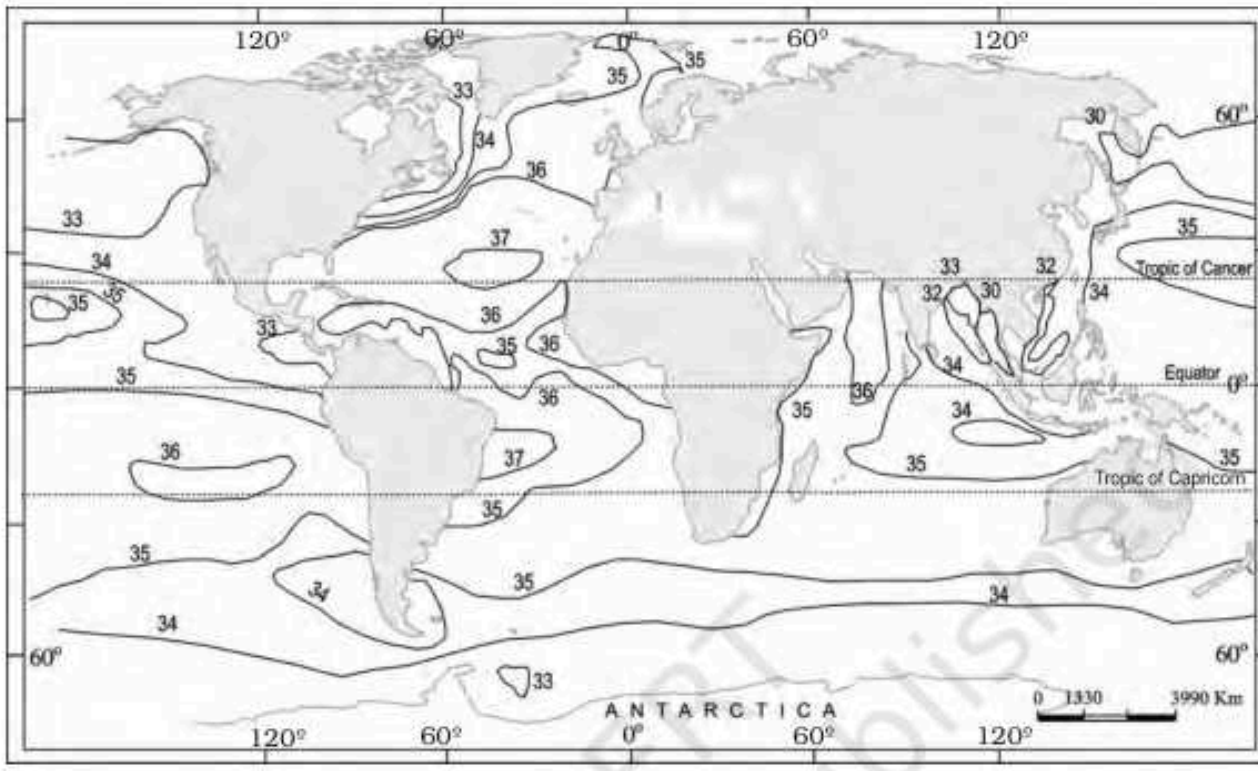


Figure 12.5 : Surface salinity of the World's Oceans

sea. Salinity at the surface increases by the loss of water to ice or evaporation, or decreased by the input of fresh waters, such as from the rivers. Salinity at depth is very much fixed, because there is no way that water is 'lost', or the salt is 'added.' There is a marked difference in the salinity between the surface zones and the deep zones of the oceans. The lower salinity water rests above the higher salinity dense water. Salinity, generally, increases with depth and there is a distinct zone called the *halocline*, where salinity increases sharply. Other factors being constant, increasing salinity of seawater causes its density to increase. High salinity seawater, generally, sinks below the lower salinity water. This leads to stratification by salinity.

EXERCISES

1. Multiple choice questions.

   (i) Identify the element which is not a part of the hydrological cycle

   (a) Evaporation (c) Precipitation

   (b) Hydration (d) Condensation

   (ii) The average depth of continental slope varies between

   (a) 2–20m (c) 20–200m

   (b) 200–2,000m (d) 2,000–20,000m

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wave moves to a new position (Figure 13.1). The actual motion of the water beneath the waves is circular. It indicates that things are carried up and forward as the wave approaches, and down and back as it passes.

Characteristics of Waves

Wave crest and trough : The highest and lowest points of a wave are called the crest and trough respectively.

Wave height : It is the vertical distance from the bottom of a trough to the top of a crest of a wave.

Wave amplitude : It is one-half of the wave height.

Wave period : It is merely the time interval between two successive wave crests or troughs as they pass a fixed point.

Wavelength : It is the horizontal distance between two successive crests.

Wave speed : It is the rate at which the wave moves through the water, and is measured in knots.

Wave frequency : It is the number of waves passing a given point during a one-second time interval.

TIDES

The periodical rise and fall of the sea level, once or twice a day, mainly due to the attraction of the sun and the moon, is called a *tide*. Movement of water caused by meteorological effects (winds and atmospheric pressure changes) are called *surges*. Surges are not regular like tides. The study of tides is very complex, spatially and temporally, as it has great variations in frequency, magnitude and height.

The moon's gravitational pull to a great extent and to a lesser extent the sun's gravitational pull, are the major causes for the occurrence of tides. Another factor is centrifugal force, which is the force that acts to counter balance the gravity. Together, the gravitational pull and the centrifugal force are responsible for creating the two major tidal bulges on the earth. On the side of the earth facing the moon, a tidal bulge occurs while on the opposite side though the gravitational attraction of the moon is less as it is farther away, the centrifugal force causes tidal bulge on the other side (Figure 13.2).

The 'tide-generating' force is the difference between these two forces; i.e. the gravitational attraction of the moon and the centrifugal force. On the surface of the earth, nearest the moon, pull or the attractive force of the moon is greater than the centrifugal force, and so there is a net force causing a bulge towards the moon. On the opposite side of the earth, the attractive force is less, as it is farther away from the moon, the centrifugal force is dominant. Hence, there is a net force away from the moon. It creates the second bulge away from the moon. On the surface of the earth, the horizontal tide generating forces are more important than the vertical forces in generating the tidal bulges.

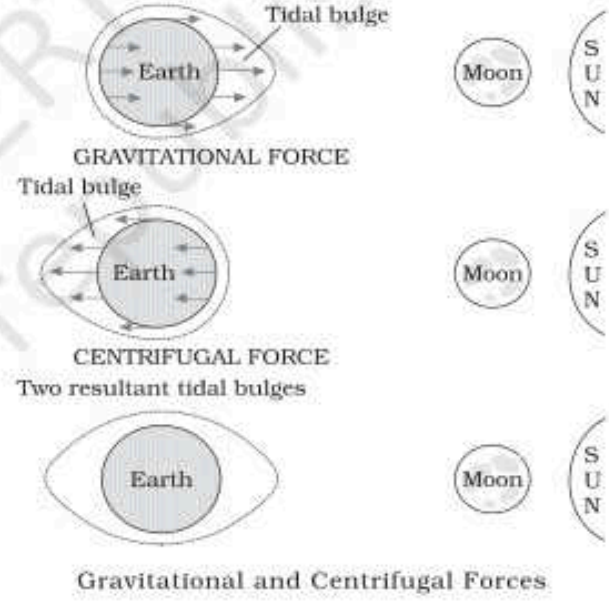


Figure 13.2 : Relation between gravitational forces and tides

The tidal bulges on wide continental shelves, have greater height. When tidal bulges hit the mid-oceanic islands they become low. The shape of bays and estuaries along a coastline can also magnify the intensity of tides. Funnel-shaped bays greatly change tidal magnitudes. When the tide is channelled between islands or into bays and estuaries they are called *tidal currents*.

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Tides of Bay of Fundy, Canada

The highest tides in the world occur in the Bay of Fundy in Nova Scotia, Canada. The tidal bulge is 15 - 16 m. Because there are two high tides and two low tides every day (roughly a 24 hour period); then a tide must come in within about a six hour period. As a rough estimate, the tide rises about 240 cm an hour (1,440 cm divided by 6 hours). If you have walked down a beach with a steep cliff alongside (which is common there), make sure you watch the tides. If you walk for about an hour and then notice that the tide is coming in, the water will be over your head before you get back to where you started!

Types of Tides

Tides vary in their frequency, direction and movement from place to place and also from time to time. Tides may be grouped into various types based on their frequency of occurrence in one day or 24 hours or based on their height.

Tides based on Frequency

Semi-diurnal tide : The most common tidal pattern, featuring two high tides and two low tides each day. The successive high or low tides are approximately of the same height.

Diurnal tide : There is only one high tide and one low tide during each day. The successive high and low tides are approximately of the same height.

Mixed tide : Tides having variations in height are known as mixed tides. These tides generally occur along the west coast of North America and on many islands of the Pacific Ocean.

Tides based on the Sun, Moon and the Earth Positions

The height of rising water (high tide) varies appreciably depending upon the position of sun and moon with respect to the earth. Spring tides and neap tides come under this category.

Spring tides : The position of both the sun and the moon in relation to the earth has direct bearing on tide height. When the sun, the moon and the earth are in a straight line, the height of the tide will be higher. These are called spring tides and they occur twice a month, one on full moon period and another during new moon period.

Neap tides : Normally, there is a seven day interval between the spring tides and neap tides. At this time the sun and moon are at right angles to each other and the forces of the sun and moon tend to counteract one another. The Moon's attraction, though more than twice as strong as the sun's, is diminished by the counteracting force of the sun's gravitational pull.

Once in a month, when the moon's orbit is closest to the earth *(perigee)*, unusually high and low tides occur. During this time the tidal range is greater than normal. Two weeks later, when the moon is farthest from earth *(apogee)*, the moon's gravitational force is limited and the tidal ranges are less than their average heights.

When the earth is closest to the sun *(perihelion)*, around 3rd January each year, tidal ranges are also much greater, with unusually high and unusually low tides. When the earth is farthest from the sun *(aphelion)*, around 4th July each year, tidal ranges are much less than average.

The time between the high tide and low tide, when the water level is falling, is called the *ebb*. The time between the low tide and high tide, when the tide is rising, is called the *flow* or *flood*.

Importance of Tides

Since tides are caused by the earth-moon-sun positions which are known accurately, the tides can be predicted well in advance. This helps the navigators and fishermen plan their activities. Tidal flows are of great importance in navigation. Tidal heights are very important, especially harbours near rivers and within estuaries having shallow 'bars' at the entrance, which prevent ships and boats from entering into the harbour. Tides are also helpful in

desilting the sediments and in removing polluted water from river estuaries. Tides are used to generate electrical power (in Canada, France, Russia, and China). A 3 MW tidal power project at Durgaduani in Sunderbans of West Bengal is under way.

Ocean Currents

Ocean currents are like river flow in oceans. They represent a regular volume of water in a definite path and direction. Ocean currents are influenced by two types of forces namely : (i) primary forces that initiate the movement of water; (ii) secondary forces that influence the currents to flow.

The primary forces that influence the currents are: (i) heating by solar energy; (ii) wind; (iii) gravity; (iv) coriolis force. Heating by solar energy causes the water to expand. That is why, near the equator the ocean water is about 8 cm higher in level than in the middle latitudes. This causes a very slight gradient and water tends to flow down the slope. Wind blowing on the surface of the ocean pushes the water to move. Friction between the wind and the water surface affects the movement of the water body in its course. Gravity tends to pull the water down the pile and create gradient variation. The Coriolis force intervenes and causes the water to move to the right in the northern hemisphere and to the left in the southern hemisphere. These large accumulations of water and the flow around them are called *Gyres*. These produce large circular currents in all the ocean basins.

Characteristics of Ocean Currents

Currents are referred to by their "drift". Usually, the currents are strongest near the surface and may attain speeds over five knots. At depths, currents are generally slow with speeds less than 0.5 knots. We refer to the speed of a current as its "drift." Drift is measured in terms of knots. The strength of a current refers to the speed of the current. A fast current is considered strong. A current is usually strongest at the surface and decreases in strength (speed) with depth. Most currents have speeds less than or equal to 5 knots.

Differences in water density affect vertical mobility of ocean currents. Water with high salinity is denser than water with low salinity and in the same way cold water is denser than warm water. Denser water tends to sink, while relatively lighter water tends to rise. Cold-water ocean currents occur when the cold water at the poles sinks and slowly moves towards the equator. Warm-water currents travel out from the equator along the surface, flowing towards the poles to replace the sinking cold water.

Types of Ocean Currents

The ocean currents may be classified based on their depth as surface currents and deep water currents : (i) *surface currents* constitute about 10 per cent of all the water in the ocean, these waters are the upper 400 m of the ocean; (ii) *deep water currents* make up the other 90 per cent of the ocean water. These waters move around the ocean basins due to variations in the density and gravity. Deep waters sink into the deep ocean basins at high latitudes, where the temperatures are cold enough to cause the density to increase.

Ocean currents can also be classified based on temperature : as cold currents and warm currents: (i) *cold currents* bring cold water into warm water areas. These currents are usually found on the west coast of the continents in the low and middle latitudes (true in both hemispheres) and on the east coast in the higher latitudes in the Northern Hemisphere; (ii) *warm currents* bring warm water into cold water areas and are usually observed on the east coast of continents in the low and middle latitudes (true in both hemispheres). In the northern hemisphere they are found on the west coasts of continents in high latitudes.

Major Ocean Currents

Major ocean currents are greatly influenced by the stresses exerted by the prevailing winds and coriolis force. The oceanic circulation pattern roughly corresponds to the earth's atmospheric circulation pattern. The air circulation over the oceans in the middle latitudes is mainly anticyclonic (more pronounced in the southern hemisphere than in the northern hemisphere). The oceanic circulation pattern also corresponds with the same. At higher latitudes,

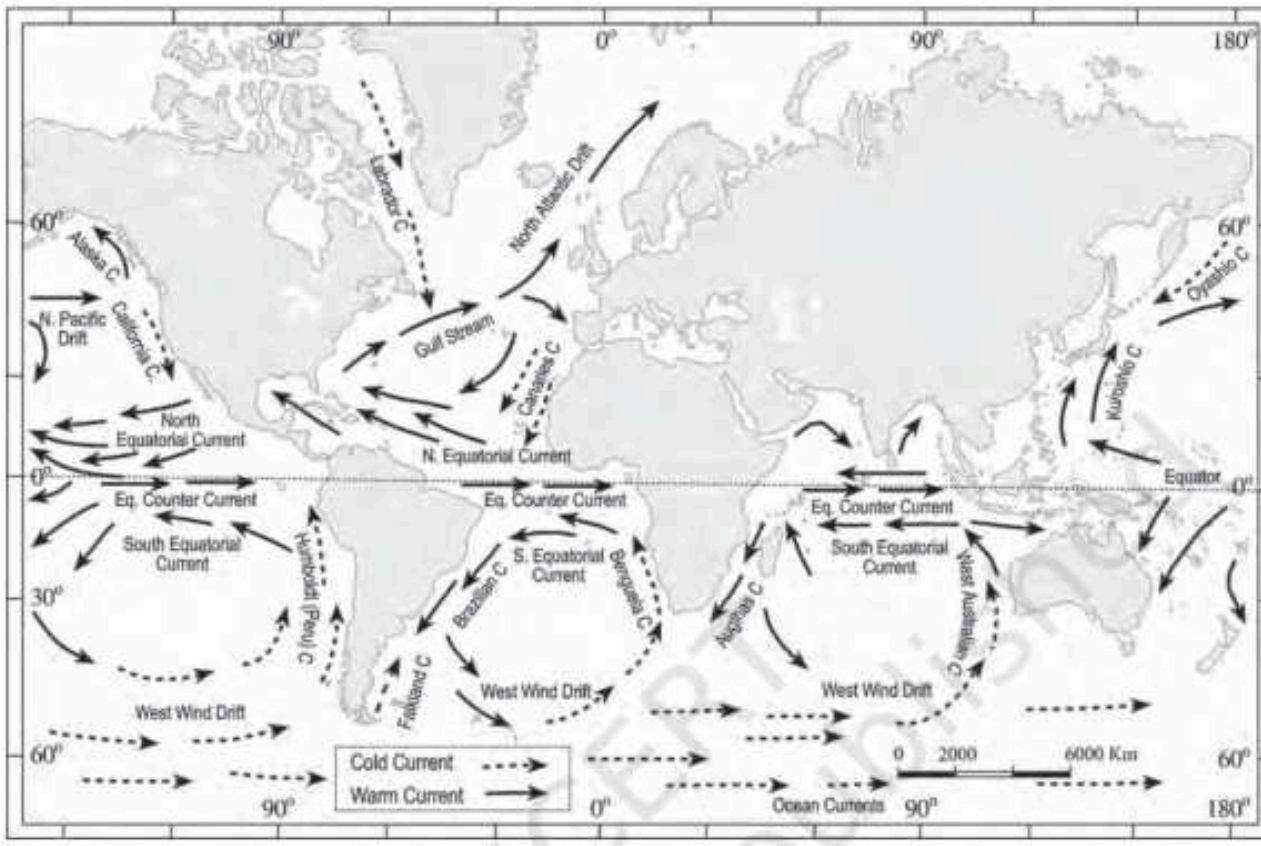


Fig.13.3 : Major currents in the Pacific, Atlantic and Indian oceans

where the wind flow is mostly cyclonic, the oceanic circulation follows this pattern. In regions of pronounced monsoonal flow, the monsoon winds influence the current movements. Due to the coriolis force, the warm currents from low latitudes tend to move to the right in the northern hemisphere and to their left in the southern hemisphere.

The oceanic circulation transports heat from one latitude belt to another in a manner similar to the heat transported by the general circulation of the atmosphere. The cold waters of the Arctic and Antarctic circles move towards warmer water in tropical and equatorial regions, while the warm waters of the lower latitudes move polewards. The major currents in the different oceans are shown in Figure13.3.

Prepare a list of currents which are found in Pacific, Atlantic and Indian Oceans.

How is the movement of currents influenced by prevailing winds? Give some examples from Figure13.3.

Effects of Ocean Currents

Ocean currents have a number of direct and indirect influences on human activities. West coasts of the continents in tropical and subtropical latitudes (except close to the equator) are bordered by cool waters. Their average temperatures are relatively low with a narrow diurnal and annual ranges. There is fog, but generally the areas are arid. West coasts of the continents in the middle and higher latitudes are bordered by warm waters which cause a distinct marine climate. They are characterised by cool summers and relatively mild winters with a narrow annual range of temperatures. Warm currents flow parallel to the east coasts of the continents in tropical and subtropical latitudes. This results in warm and rainy climates. These areas lie in the western margins of the subtropical anti-cyclones. The mixing of warm and cold currents help to replenish the oxygen and favour the growth of planktons, the primary food for fish population. The best fishing grounds of the world exist mainly in these mixing zones.

UNIT VI

LIFE ON THE EARTH

This unit deals with

- *Biosphere — biodiversity and conservation*

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CHAPTER

14

BIODIVERSITY AND CONSERVATION

You have already learnt about the geomorphic processes particularly weathering and depth of weathering mantle in different climatic zones. See the Figure 5.2 in Chapter 5 in order to recapitulate. You should know that this weathering mantle is the basis for the diversity of vegetation and hence, the biodiversity. The basic cause for such weathering variations and resultant biodiversity is the input of solar energy and water. No wonder that the areas that are rich in these inputs are the areas of wide spectrum of biodiversity.

> Biodiversity as we have today is the result of 2.5-3.5 billion years of evolution. Before the advent of humans, our earth supported more biodiversity than in any other period. Since, the emergence of humans, however, biodiversity has begun a rapid decline, with one species after another bearing the brunt of extinction due to overuse. The number of species globally vary from 2 million to 100 million, with 10 million being the best estimate. New species are regularly discovered most of which are yet to be classified (an estimate states that about 40 per cent of fresh water fishes from South America are not classified yet). Tropical forests are very rich in bio-diversity.

Biodiversity is a system in constant evolution, from a view point of species, as well as from view point of an individual organism. The average half-life of a species is estimated at between one and four million years, and 99 per cent of the species that have ever lived on the earth are today extinct. Biodiversity is not found evenly on the earth. It is consistently richer in the tropics. As one approaches the polar regions, one finds larger and larger populations of fewer and fewer species.

Biodiversity itself is a combination of two words, *Bio* (life) and *diversity* (variety). In simple terms, biodiversity is the number and variety of organisms found within a specified geographic region. It refers to the varieties of plants, animals and micro-organisms, the genes they contain and the ecosystems they form. It relates to the variability among living organisms on the earth, including the variability within and between the species and that within and between the ecosystems. Biodiversity is our living wealth. It is a result of hundreds of millions of years of evolutionary history.

Biodiversity can be discussed at three levels : (i) Genetic diversity; (ii) Species diversity; (iii) Ecosystem diversity.

Genetic Diversity

Genes are the basic building blocks of various life forms. Genetic biodiversity refers to the variation of genes within species. Groups of individual organisms having certain similarities in their physical characteristics are called *species*. Human beings genetically belong to the *homo sapiens* group and also differ in their characteristics such as height, colour, physical appearance, etc., considerably. This is due to genetic diversity. This genetic diversity is essential for a healthy breeding of population of species.

Species Diversity

This refers to the variety of species. It relates to the number of species in a defined area. The diversity of species can be measured through its richness, abundance and types. Some areas are more rich in species than others. Areas rich in species diversity are called *hotspots* of diversity (Figure 14.5).

Ecosystem Diversity

You have studied about the ecosystem in the earlier chapter. The broad differences between ecosystem types and the diversity of habitats and ecological processes occurring within each ecosystem type constitute the ecosystem diversity. The 'boundaries' of communities (associations of species) and ecosystems are not very rigidly defined. Thus, the demarcation of ecosystem boundaries is difficult and complex.



Figure 14.1 : Grasslands and sholas in Indira Gandhi National Park, Annamalai, Western Ghats — an example of ecosystem diversity

Importance of Biodiversity

Biodiversity has contributed in many ways to the development of human culture and, in turn, human communities have played a major role in shaping the diversity of nature at the genetic, species and ecological levels. Biodiversity plays the following roles: ecological, economic and scientific.

Ecological Role of Biodiversity

Species of many kinds perform some function or the other in an ecosystem. Nothing in an ecosystem evolves and sustains without any reason. That means, every organism, besides extracting its needs, also contributes something of useful to other organisms. Can you think of the way we, humans contribute to the sustenance of ecosystems. Species capture and store energy, produce and decompose organic materials, help to cycle water and nutrients throughout the ecosystem, fix atmospheric gases and help regulate the climate. These functions are important for ecosystem function and human survival. The more diverse an ecosystem, better are the chances for the species to survive through adversities and attacks, and consequently, is more productive. Hence, the loss of species would decrease the ability of the system to maintain itself. Just like a species with a high genetic diversity, an ecosystem with high biodiversity may have a greater chance of adapting to environmental change. In other words, the more the variety of species in an ecosystem, the more stable the ecosystem is likely to be.

Economic Role of Biodiversity

For all humans, biodiversity is an important resource in their day-to-day life. One important part of biodiversity is 'crop diversity', which is also called agro-biodiversity. Biodiversity is seen as a reservoir of resources to be drawn upon for the manufacture of food, pharmaceutical, and cosmetic products. This concept of biological resources is responsible for the deterioration of biodiversity. At the same time, it is also the origin of new conflicts dealing with rules of division and appropriation of natural resources. Some of the important economic commodities that biodiversity supplies to humankind are: food crops, livestock, forests, fish, medicinal resources, etc.

Scientific Role of Biodiversity

Biodiversity is important because each species can give us some clue as to how life evolved and will continue to evolve. Biodiversity also helps in understanding how life functions and the role of each species in sustaining

ecosystems of which we are also a species. This fact must be drawn upon every one of us so that we live and let other species also live their lives.

It is our ethical responsibility to consider that each and every species along with us have an intrinsic right to exist. Hence, it is morally wrong to voluntarily cause the extinction of any species. The level of biodiversity is a good indicator of the state of our relationships with other living species. In fact, the concept of biodiversity is an integral part of many human cultures.

Loss of Biodiversity

Since the last few decades, growth in human population has increased the rate of consumption of natural resources. It has accelerated the loss of species and habitation in different parts of the world. Tropical regions which occupy only about one-fourth of the total area of the world, contain about three-fourth of the world human population. Over-exploitation of resources and deforestation have become rampant to fulfil the needs of large population. As these tropical rain forests contain 50 per cent of the species on the earth, destruction of natural habitats have proved disastrous for the entire biosphere.

Natural calamities such as earthquakes, floods, volcanic eruptions, forest fires, droughts, etc. cause damage to the flora and fauna of the earth, bringing change the biodiversity of respective affected regions. Pesticides and other pollutants such as hydrocarbons and toxic heavy metals destroy the weak and sensitive species. Species which are not the natural inhabitants of the local habitat but are introduced into the system, are called *exotic species*. There are many examples when a natural biotic community of the ecosystem suffered extensive damage because of the introduction of exotic species. During the last few decades, some animals like tigers, elephants, rhinoceros, crocodiles, minks and birds were hunted mercilessly by poachers for their horn, tusks, hides, etc. It has resulted in the rendering of certain types of organisms as endangered category.

The International Union of Conservation of Nature and Natural Resources (IUCN) has classified the threatened species of plants and animals into three categories for the purpose of their conservation.

Endangered Species

It includes those species which are in danger of extinction. The IUCN publishes information about endangered species world-wide as the *Red List* of threatened species.



Figure 14.2 : Red Panda — an endangered species

Vulnerable Species

This includes the species which are likely to be in danger of extinction in near future if the factors threatening to their extinction continue. Survival of these species is not assured as their population has reduced greatly.

Rare Species

Population of these species is very small in the world; they are confined to limited areas or thinly scattered over a wider area.

Conservation of Biodiversity

Biodiversity is important for human existence. All forms of life are so closely interlinked that disturbance in one gives rise to imbalance in the others. If species of plants and animals become endangered, they cause degradation in the environment, which may threaten human being's own existence.



Figure 14.3 : Humbodtia decurrens Bedd — highly rare endemic tree of Southern Western Ghats (India)

There is an urgent need to educate people to adopt environment-friendly practices and reorient their activities in such a way that our development is harmonious with other life forms and is sustainable. There is an increasing consciousness of the fact that such conservation with sustainable use is possible only with the involvement and cooperation of local communities and individuals. For this, the development of institutional structures at local levels is necessary. The critical problem is not merely the conservation of species nor the habitat but the continuation of process of conservation.

The Government of India along with 155 other nations have signed the Convention of Biodiversity at the Earth Summit held at Rio de Janeiro, Brazil in June 1992. The world conservation strategy has suggested the following steps for biodiversity conservation:

(i) Efforts should be made to preserve the species that are endangered.
(ii) Prevention of extinction requires proper planning and management.
(iii) Varieties of food crops, forage plants, timber trees, livestock, animals and their wild relatives should be preserved;
(iv) Each country should identify habitats of wild relatives and ensure their protection.
(v) Habitats where species feed, breed, rest and nurse their young should be safeguarded and protected.
(vi) International trade in wild plants and animals be regulated.

To protect, preserve and propagate the variety of species within natural boundaries, the Government of India passed the *Wild Life (Protection) Act, 1972*, under which national parks and sanctuaries were established and biosphere reserves declared. Details of these biosphere reserves are given in the book *India: Physical Environment* (NCERT, 2006).

There are some countries which are situated in the tropical region; they possess a large number of the world's species diversity. They are called *mega diversity centres*. There are 12 such countries, namely Mexico, Columbia, Ecuador, Peru, Brazil, Democratic Republic of Congo, Madagascar, China, India, Malaysia, Indonesia and Australia in which these centres are located. In order to concentrate resources on those areas that are most vulnerable, the International Union for the Conservation of Nature and Natural Resources (IUCN) has identified certain areas as biodiversity hotspots (Figure 14.1). Hotspots are defined according to their vegetation. Plants are important because these determine the primary productivity of an ecosystem. Most, but not all, of the hotspots rely on species-rich ecosystems for food, firewood, cropland, and income from timber. In Madagascar, for example, about 85 per cent of the plants and animals are found nowhere else in the world, Other hotspots in wealthy countries are facing different types of pressures. The islands of Hawaii have many unique plants and animals that are threatened by introduced species and land development.

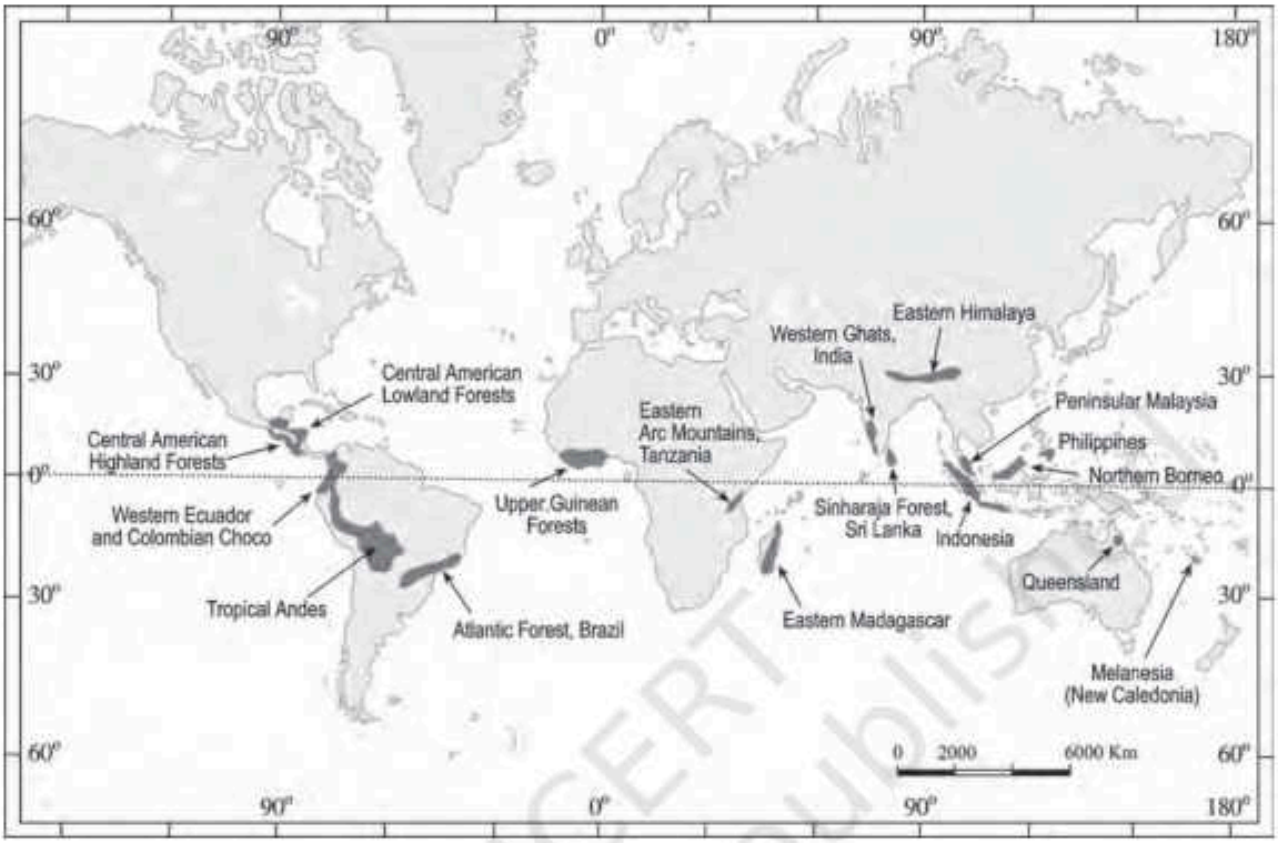


Figure 14.4 : Some ecological 'hotspots' in the world

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EXERCISES

1. Multiple choice questions.

   (i) Conservation of biodiversity is important for :

   (a) Animals (c) Plants

   (b) Animals and plants (d) All organisms

   (ii) Threatened species are those which :

   (a) threaten others

   (b) Lion and tiger

   (c) are abundant in number

   (d) are suffering from the danger of extinction

   (iii) National parks and sanctuaries are established for the purpose of :

   (a) Recreation (c) Pets

   (b) Hunting (d) Conservation

(iv) Biodiversity is richer in :

(a) Tropical Regions (c) Temperate Regions

(b) Polar Regions (d) Oceans

(v) In which one of the following countries, the 'Earth Summit' was held?

(a) the UK (c) Brazil

(b) Mexico (d) China

2. Answer the following questions in about 30 words.

(i) What is biodiversity?

(ii) What are the different levels of biodiversity?

(iii) What do you understand by 'hotspots'?

(iv) Discuss briefly the importance of animals to human kind.

(v) What do you understand by 'exotic species'?

3. Answer the following questions in about 150 words.

(i) What are the roles played by biodiversity in the shaping of nature?

(ii) What are the major factors that are responsible for the loss of biodiversity? What steps are needed to prevent them?

Project Work

Collect the names of national parks, sanctuaries and biosphere reserves of the state where your school is located and show their location on the map of India.

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Chapter 6

Introduction To Remote Sensing

Both the human eyes and the photographic systems respond to light in a minute portion of the total energy received and responded by the objects' surface. The present day remote sensing devices, on the other hand, react to much wider range of radiations reflected/ emitted, absorbed and transmitted by all object surfaces at a temperature above 0 Kelvin (-273 C).

The term remote sensing was first used in the early 1960s. Later, it was defined as the total processes used to acquire and measure the information of some property of objects and phenomena by a recording device (sensor) that is not in physical contact with the objects and phenomena in study. It can be noted from the above definition of remote sensing that it primarily involves an object surface, the recording device and the information carrying energy waves (Fig 6.1).



Figure 6.1 Conceptual Frame of Remote Sensing

Glossary

Absorptance : The ratio of the radiant energy absorbed by a substance to the energy it receives.

Band : The specific wavelength interval in the electromagnetic spectrum.

Digital image : An array of digital numbers (DN) arranged in rows and columns, having the property of an intensity value and their locations.

Digital Number : An intensity value of a pixel in a digital image.

Digital Image Processing : The numerical manipulation of DN values for the purpose of extracting information about the phenomena of the surface they represent.

Electromagnetic Radiation (EMR) : The Energy propagated through a space or a medium at a speed of light.

Electromagnetic Spectrum : The continuum of EMR that ranges from short wave high frequency cosmic radiations to long wavelength low frequency radio waves.

False Colour Composite (FCC) : An artificially generated colour image in which blue, green and red colours are assigned to the wavelength regions to which they do not belong in nature. For example, in standard a False Colour Composite blue is assigned to green radiations (0.5 to 0.6 µm), green is assigned to red radiations (0.6 to 0.7 µm and red is assigned to Near Infrared radiation (0.7 to 0.8 µm).

Gray scale : A medium to calibrate the variations in the brightness of an image that ranges from black to white with intermediate grey values.

Image : The permanent record of a scene comprising of natural and man-made features and activities, produced by photographic and non-photographic means.

Scene : The ground area covered by an image or a photograph.

Sensor : Any imaging or non-imaging device that receives EMR and converts it into a signal that can be recorded and displayed as photographic or digital image.

Reflectance : The ratio of the radiant energy reflected by a substance to the energy it receives.

Spectral Band : The range of the wavelengths in the continuous spectrum such as the green band ranges from 0.5 to .6 µ and the range of NIR band 0.7 to 1.1 µ.

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Stages in Remote Sensing

Figure 6.2 illustrates the processes used in remote sensing data acquisition. These basic processes that help in the collection of information about the properties of the objects and phenomena of the earth surface are as follows :

(a) Source of Energy (sun/self-emission);
(b) Transmission of energy from the source to the surface of the earth;
(c) Interaction of energy with the earth's surface;
(d) Propagation of reflected/emitted energy through atmosphere;
(e) Detection of the reflected/emitted energy by the sensor;
(f) Conversion of energy received into photographic/digital form of data;
(g) Extraction of the information contents from the data products; and
(h) Conversion of information into Map/Tabular forms.

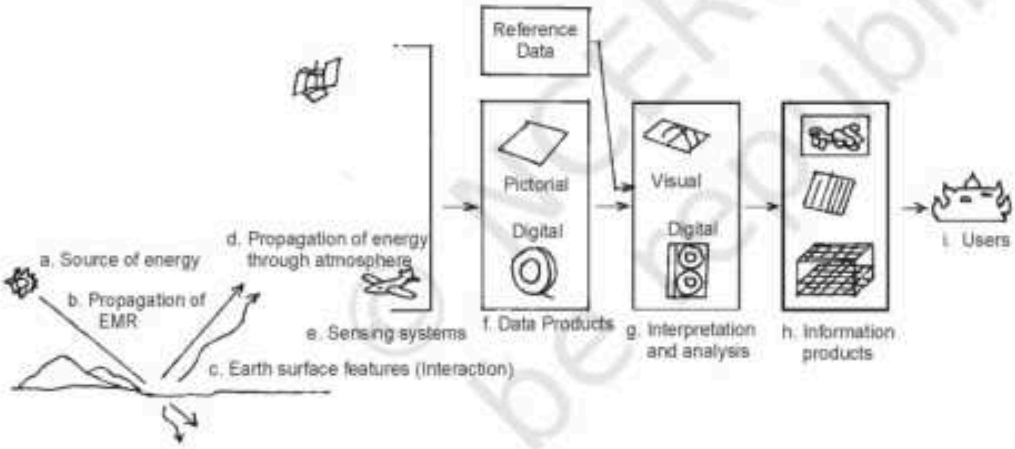


Figure 6.2 Stages in Remote Sensing Data Acquisition

a. Source of Energy: Sun is the most important source of energy used in remote sensing. The energy may also be artificially generated and used to collect information about the objects and phenomena such as flashguns or energy beams used in radar (radio detection and ranging).

b. Transmission of Energy from the Source to the Surface of the Earth: The energy that emanates from a source propagates between the source and the object surface in the form of the waves of

energy at a speed of light (300,000 km per second). Such energy propagation is called the *Electromagnetic Radiation* (EMR). The energy waves vary in size and frequency. The plotting of such variations is known as the *Electromagnetic Spectrum* (Fig. 6.3). On the basis of the size of the waves and frequency, the energy waves are grouped into Gamma, X-rays, Ultraviolet rays, Visible rays, Infrared rays , Microwaves and Radio waves. Each one of these broad regions of spectrum is used in different applications. However, the visible, infrared and microwave regions of energy are used in remote sensing.

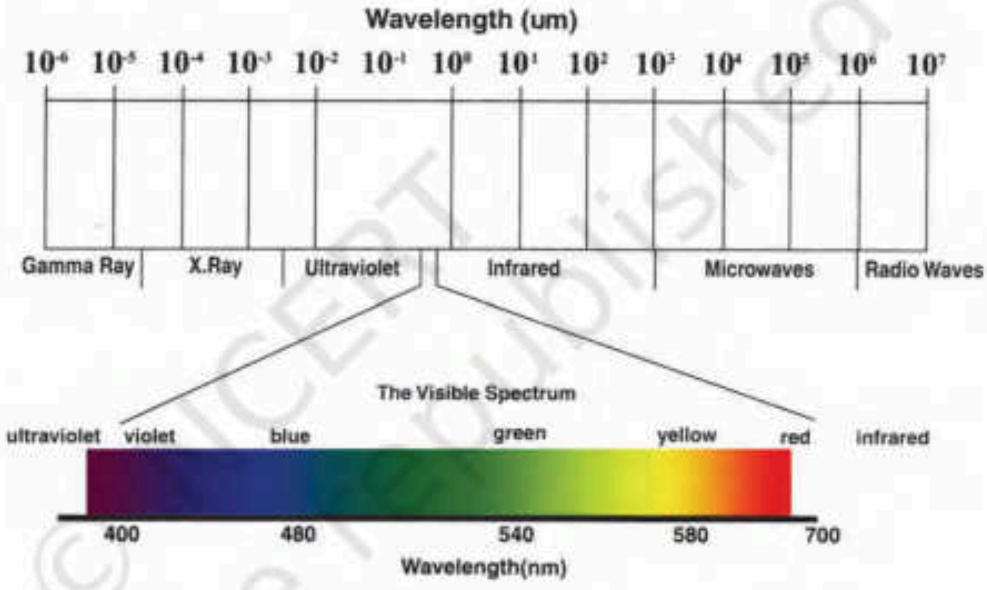


Figure 6.3 Electromagnetic Spectrum

c. Interaction of Energy with the Earth's Surface: The propagating energy finally interacts with the objects of the surface of the earth. This leads to absorption, transmission, reflection or emission of energy from the objects. We all know that all objects vary in their composition, appearance forms and other properties. Hence, the objects' responses to the energy they receive are also not uniform. Besides, one particular object also responds differently to the energy it receives in different regions of the spectrum (Fig. 6.5). For example, a fresh water body absorbs more energy in the red and infrared regions of the spectrum and appears dark/black in a satellite image whereas turbid water body reflects more in blue and green regions of spectrum and appears in light tone (Fig. 6.4).

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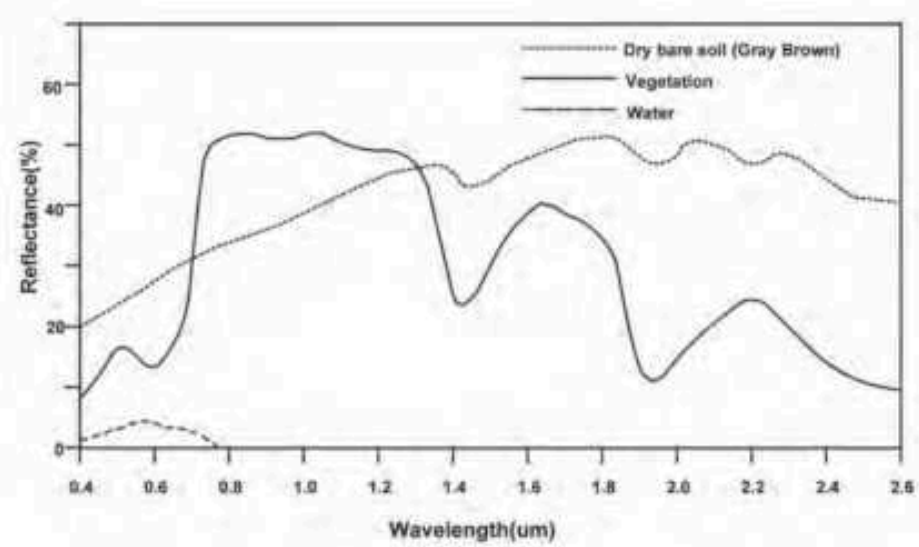


Figure 6.4 Spectral Signature of Soil, Vegetation and Water

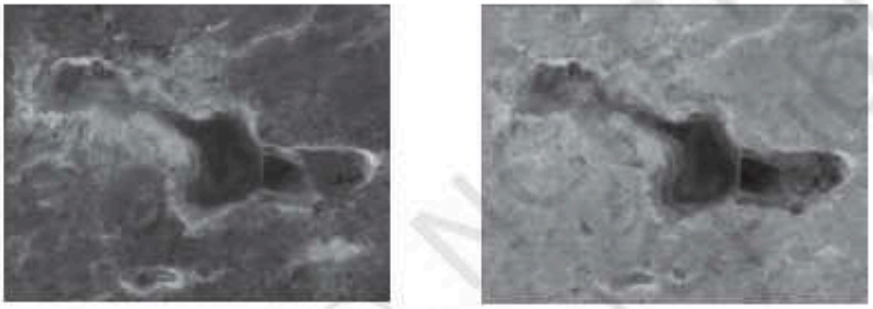


Figure 6.5 IRS 1 C Band 1 Green (Left) and Band 4 IR Images of Sambhar Lake, Rajasthan

d. Propagation of Reflected/Emitted Energy through Atmosphere: When energy is reflected from objects of the earth's surface, it re–enters into the atmosphere. You may be aware of the fact that atmosphere comprises of gases, water molecules and dust particles. The energy reflected from the objects comes in contact with the atmospheric constituents and the properties of the original energy get modified. Whereas the Carbon dioxide (CO_2) the Hydrogen (H), and the water molecules absorb energy in the middle infrared region, the dust particles scatter the blue energy. Hence, the energy that is either absorbed or scattered by the atmospheric constituents never reaches to sensor placed onboard a satellite and the properties of the objects carried by such energy waves are left unrecorded.

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e. Detection of Reflected/Emitted Energy by the Sensor:

The sensors recording the energy that they receive are placed in a near–polar sun synchronous orbit at an altitude of 700 – 900 km. These satellites are known as remote sensing satellites (e.g. Indian Remote Sensing Series). As against these satellites, the weather monitoring and telecommunication satellites are placed in a Geostationary position (the satellite is always positioned over its orbit that synchronises with the direction of the rotation of the earth) and revolves around the earth (coinciding with the direction of the movement of the earth over its axis) at an altitude of nearly 36,000 km (e.g. INSAT series of satellites). A comparison between the remote sensing and weather monitoring satellites is given in Box (6.1). Figure 6.6 shows the orbits of Sun Synchronous and Geostationary satellites respectively.

Box. 6.1 Comparison between Sun-Synchronous and Geostationary Satellites

Orbital Characteristics	Sun Synchronous Satellites	Geostationary Satellites
Altitude	700 – 900 km	@ 36,000 km
Coverage	81°N to 81°S	1/3rd of the Globe
Orbital period	@ 14 orbits per day	24 hours
Resolution	Fine (182 metre to 1 metre)	Coarse (1 km x 1 km)
Uses	Earth Resources Applications	Telecommunication and Weather monitoring

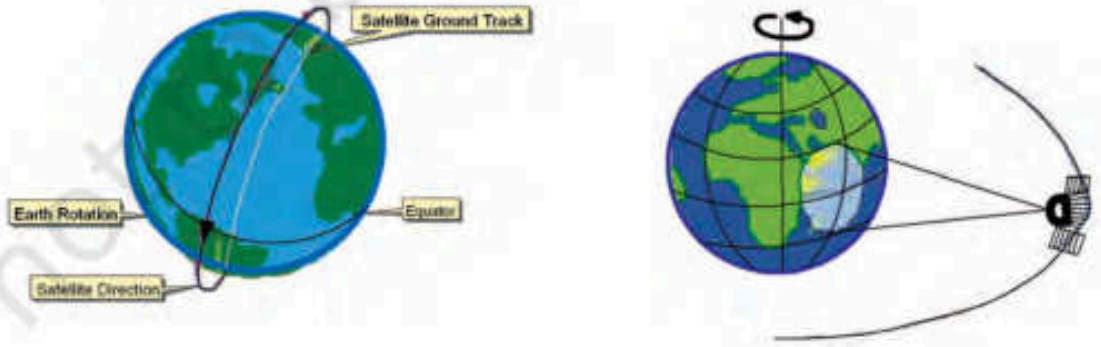


Figure 6.6 Orbit of Sun Synchronous (Left) and Geostationary (Right) Satellites

Remote sensing satellites are deployed with sensors which are capable of collecting the EMR reflected by the objects. Photographic camera obtains

photographs at an instance of exposure. However, the sensors used in remote sensing satellites possess a mechanism that is different from photographic camera in collecting and recording the information. The images so acquired by space-borne sensors are in digital format as against the photographic format obtained through a camera-based system.

f. Conversion of Energy Received into Photographic/ Digital Form of Data: The radiations received by the sensor are electronically converted into a digital image. It comprises digital numbers that are arranged in rows and columns. These numbers may also be converted into an analogue (picture) form of data product. The sensor onboard an earth-orbiting satellite electronically transmits the collected image data to an Earth Receiving Station located in different parts of the world. In India, one such station is located at Shadnagar near Hyderabad.

g. Extraction of Information Contents from Data Products: After the image data is received at the earth station, it is processed for elimination of errors caused during image data collection. Once the image is corrected, information extraction is carried out from digital images using digital image processing techniques and from analogue form of data products by applying visual interpretation methods.

h. Conversion of Information into Map/Tabular Forms: The interpreted information is finally delineated and converted into different layers of thematic maps. Besides, quantitative measures are also taken to generate a tabular data.

Sensors

A sensor is a device that gathers electromagnetic radiations, converts it into a signal and presents it in a form suitable for obtaining information about the objects under investigation. Based upon the form of the data output, the sensors are classified into photographic (analogue) and non–photographic (digital) sensors.

A photographic sensor (camera) records the images of the objects at an instance of exposure. On the other hand, a non–photographic sensor obtains the images of the objects in bit-by-bit form. These sensors are known as *scanners*. In the present chapter, we will confine ourselves to describe the non–photographic sensors that are used in satellite remote sensing.

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Multispectral Scanners: In satellite remote sensing, the Multi Spectral Scanners (MSS) are used as sensors. These sensors are designed to obtain images of the objects while sweeping across the field of view. A scanner is usually made up of a reception system consisting of a mirror and detectors. A scanning sensor constructs the scene by recording a series of scan lines. While doing so, the motor device oscillates the scanning mirror through the angular field of view of the sensor, which determines the length of scan lines and is called *swath*. It is because of such reasons that the mode of collection of images by scanners is referred bit–by–bit. Each scene is composed of cells that determine the spatial resolution of an image. The oscillation of the scanning mirror across the scene directs the received energy to the detectors, where it is converted into electrical signals. These signals are further converted into numerical values called *Digital Number* (DN Values) for recording on a magnetic tape.

The Multi-Spectral Scanners are divided into the following types:

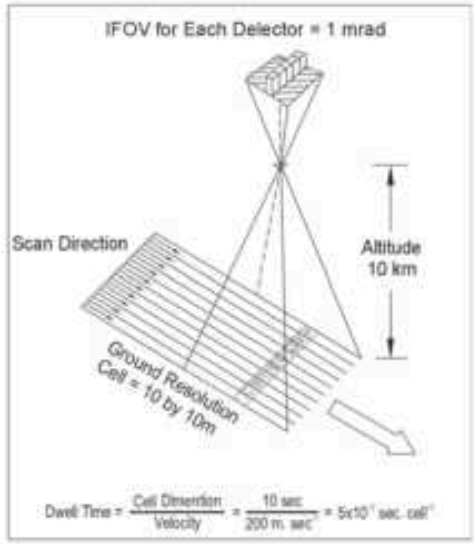
(i) Whiskbroom Scanners

(ii) Pushbroom Scanners

(i) Whiskbroom Scanners : The whiskbroom scanners are made up of a rotating mirror and a single detector. The mirror is so oriented that when it completes a rotation, the detector sweeps across the field of view



6.7 Whiskbroom Scanners



6.8 Pushbroom Scanners

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between 90 and 120 to obtain images in a large number of narrow spectral bands ranging from visible to middle infrared regions of the spectrum. The total extent of the oscillating sensor is known as the Total Field of View (TFOV) of the scanner. While scanning the entire field, the sensor's optical head is always placed at a particular dimension called the Instantaneous Field of View (IFOV). Figure 6.7 depicts the scanning mechanism of whiskbroom scanners.

(i) Pushbroom Scanners: The pushbroom scanners consist of a number of detectors which are equivalent to the number obtained by dividing the *swath* of the sensor by the size of the spatial resolution (Fig. 6.8). For example, the *swath* of High Resolution Visible Radiometer – 1 (HRVR – 1) of the French remote sensing satellite SPOT is 60 km and the spatial resolution is 20 metres. If we divide 60 km x 1000 metres/20 metres, we get a number of 3000 detectors that are deployed in SPOT HRV – 1 sensor. In pushbroom scanner, all detectors are linearly arrayed and each detector collects the energy reflected by the ground cell (pixel) dimensions of 20 metres at a nadir's view.

Resolving Powers of the Satellites

In satellite remote sensing, the sun-synchronous polar orbit enables the collection of images after a pre-determined periodical interval referred to as the temporal resolution or the revisit time of the satellite over the same area of the earth surface. Fig. 6.9 illustrates the two images acquired over two different periods in time for the same area enabling to study and record the changes that take place with respect to the types of vegetation in Himalayas. In another example, Fig. 6.10 (a and b) shows the images acquired before and after the tsunami in the Indian Ocean. The image acquired in June 2004 clearly shows the undisturbed topography of Banda Aceh in Indonesia, whereas the post tsunami image acquired immediately after tsunami reveals the damages that were caused by the tsunami.

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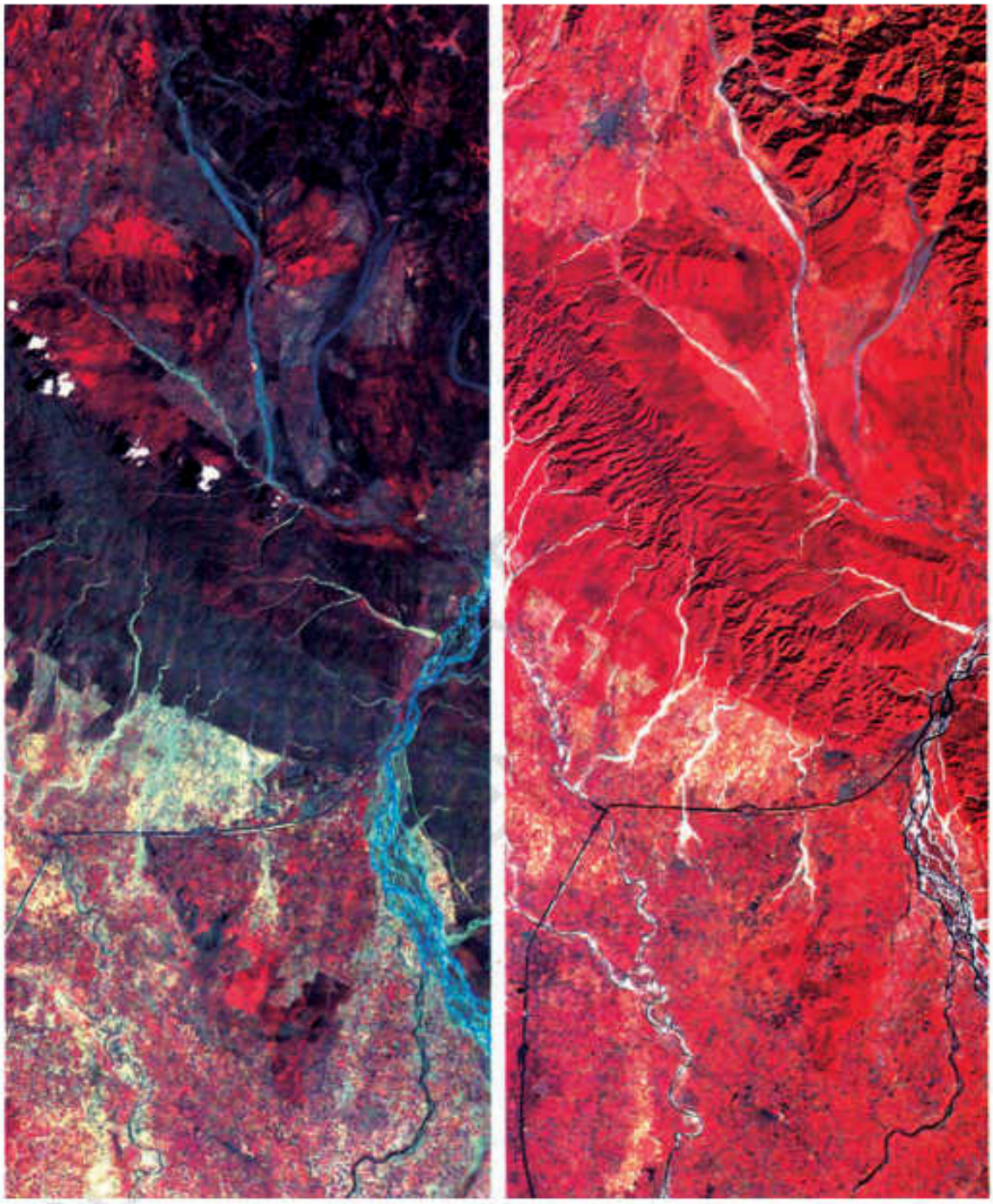


Figure 6. 9 Images of Himalayas and Northern Indian Plain by IRS Satellite taken in May (Left) and November (Right) show differences in the types of vegetation. The red patches in May image refer to Coniferous vegetation. In November image the additional red patches refer to Deciduous plants and the light red colour is related to the crops.



Figure 6.10 (a) Pre-tsunami Image acquired in June 2004



Figure 6.10 (b) Post-tsunami image acquired in December, 2004

Sensor Resolutions

Remote sensors are characterised by spatial, spectral and radiometric resolutions that enable the extraction of useful information pertaining to different terrain conditions.

(i) Spatial Resolution: You must have seen some people using spectacles while reading a book or newspaper. Have you ever thought as to why they do so. It is simply because of the fact that resolving power of their eyes to differentiate two closed spaced letters in a word is unable to identify them as two different letters. By using positive spectacles they try to improve their vision as well as the resolving power. In remote sensing, the spatial resolution of the sensors refers to the same phenomena. It is the capability of the sensor to distinguish two closed spaced object surfaces as two different object surfaces. As a rule, with an increasing resolution the identification of even smaller object surfaces become possible.

(ii) Spectral Resolution: It refers to the sensing and recording power of the sensor in different bands of EMR (Electromagnetic radiation). Multispectral images are acquired by using a device that disperses the radiation received by the sensor and recording it by deploying detectors sensitive to specific spectral ranges. The principles in obtaining such images is the extension of the dispersion of light in nature resulting in the appearance of the 'rainbow" and the use of prism in the lab (Box 6.2).

The images obtained in different bands show objects response differently as discussed in Para 3 of the stages in remote sensing data acquisition. Fig. 6.11 illustrates images acquired in different spectral regions by IRS P - 6 (Resource sat - 1) showing strong absorption properties of fresh water in band 4 (Infrared) and mixed strong reflectance in band 2 (green) by dry surfaces (Fig. 6.11).

(iii) Radiometric Resolution: It is the capability of the sensor to discriminate between two targets. Higher the radiometric resolution, smaller the radiance differences that can be detected between two targets.

The spatial, spectral, and radiometric resolutions of some of the remote sensing satellites of the world are shown in Table 6.1.

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trajectories of other nations such as the USA and Germany, two significant industrial powers. It also displayed the human and material costs incurred by Britain on its industrialisation – the plight of the labouring poor, especially of children, environmental degradation and the consequent epidemics of cholera and tuberculosis. In Theme 7 you will read how industrial pollution and cadmium and mercury poisoning in Japan stirred people into mass movements against indiscriminate industrialisation.

European powers began to colonise parts of America and Asia and South Africa well before the Industrial Revolution. Theme 6 tells you the story of what European settlers did to the native peoples of America and Australia. The bourgeois mentality of the settlers made them buy and sell everything, including land and water. But the natives, who appeared uncivilised to European Americans, asked, 'If you do not own the freshness of the air and the sparkle of the water, how can one buy them?' The natives did not feel the need to own land, fish or animals. They had no desire to *commodify* them; if things needed to be exchanged, they could simply be *gifted*. Quite obviously, the natives and the Europeans represented competing notions of civilisation. The former did not allow the European deluge to wipe out their cultures although the US and Canadian governments of the mid-twentieth century desired natives to 'join the mainstream' and the Australian authorities of the same period attempted to simply ignore their traditions and culture. One might wonder what is meant by 'mainstream'. How does economic and political power influence the making of 'mainstream cultures'?

Western capitalisms – mercantile, industrial and financial – and early twentieth-century Japanese capitalism created colonies in large parts of the third world. Some of these were settler colonies. Others, such as British rule in India, are examples of direct imperial control. The case of nineteenth- and early twentieth-century China illustrates a third variant of imperialism. Here Britain, France, Germany, Russia, America and Japan meddled in Chinese affairs without directly taking over state power. They exploited the country's resources to their own advantage, seriously compromising Chinese sovereignty and reducing the country to the status of a semi-colony.

Almost everywhere, colonial exploitation was challenged by powerful nationalist movements. Nationalisms, however, also arose without a colonial context, as in the West or Japan. All



Linking the world – J. Lipchitz's Figure, *sculpted in the 1920s, shows the influence of central African statuary.*

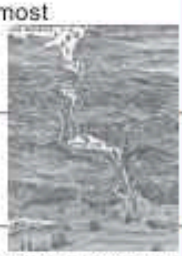
Linking the world – Japanese Zen paintings like this one were admired by western artists, and influenced the 'Abstract Expressionist' style of painting in the 1920s in the USA.



nationalisms are doctrines of popular sovereignty. Nationalist movements believe that political power should rest with the people and this is what makes nationalism a modern concept. Civic nationalism vests sovereignty in *all* people regardless of language, ethnicity, religion or gender. It seeks to create a community of rights-exercising citizens and defines nationhood in terms of *citizenship*, not ethnicity or religion. Ethnic and religious nationalisms try to build national solidarities around a given language, religion or set of traditions, defining the people ethnically, not in terms of common citizenship. In a multi-ethnic country, ethnic nationalists might limit the exercise of sovereignty to a chosen people, often assumed to be superior to minority communities. Today, most western countries define their nationhood in terms of common citizenship and not by common ethnicity. One prominent exception is Germany where ideas of ethnic nationalism have had a long and troubling career going back to the reaction against the French imperial occupation of German states in 1806. Ideologies of civic nationalism have vied with those of ethnic/religious nationalism the world over and this has been so in modern India, China and Japan as well.

As with industrialisation, so with paths to modernisation. Different societies have evolved their distinctive modernities. The Japanese and Chinese cases, along with the stories of Taiwan and South Korea, are very instructive in this regard. Japan succeeded in remaining free of colonial control and achieved fairly rapid economic and industrial progress throughout the twentieth century. The rebuilding of the Japanese economy after a humiliating defeat in the Second World War should not be seen as a mere post-war miracle. As Theme 7 shows, it resulted from certain gains that had already been accomplished in the nineteenth and early twentieth centuries. Did you know, for instance, that by 1910 tuition fees for studying at a primary school had more or less ended and enrolment had become universal? Japan's path to modernisation, like that of any other country, has had its own tensions: those between democracy and militarism, ethnic nationalism and civic nation-building and between what many Japanese describe as 'tradition' and 'westernisation'.

The Chinese resisted colonial exploitation and their own bureaucratic landed elite through a combination of peasant rebellion, reform and revolution. By the early 1930s, the Chinese Communist Party, which drew its strength from

DATES	AFRICA	EUROPE
1860-70	Suez Canal*, one of the most important trade routes in the world, opens (1869) 	Russian serfs are freed (1861)
1870-80		Germany and Italy emerge as unified nation-states
1880-90	Beginning of the European 'Scramble for Africa'	
1890-1900		Making of the first film (1895); the modern Olympics are held for the first time in Athens (1896)
1900-1910	Mahatma Gandhi* advocates satyagraha to resist racist laws (1906)	
1910-1920	South Africa introduces laws to reserve 87 per cent of land for whites (1913)	First World War (1914-1918); the Russian Revolution of 1917
1920-30		Turkey becomes a republic under Mustapha Kemal (1923)
1930-40	First trans-African railway from Angola to Mozambique completed (1931)	Hitler captures power in Germany (1933); Second World War (1939-45)
1940-50	Afrikaner National Party wins power in South Africa (1948). The policy of Apartheid is put in place	Britain recognises Irish independence (1949)
1950-60	Ghana is the first country in sub-Saharan Africa to become independent (1957)	Discovery of DNA; Russia launches the spacecraft Sputnik (1957)
1960-70	Organisation of African Unity founded (1963)	Protest movements in Europe (1968)
1970-80		
1980-90		Mikhail Gorbachev, leader of the USSR (1985); Beginning of the world wide web (1989)
1990-2000	Nelson Mandela* freed in South Africa (1990); process of dismantling Apartheid begins	Scientists clone the sheep Dolly (1997) raising new debates about the limits of genetic engineering

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DATES	ASIA	SOUTH ASIA
1720-30	*Gujin tushu jicheng**, the largest encyclopaedia ever printed, commissioned by Kangxi, the Manchu ruler of China	
1730-40		
1740-50		Marathas extend control over northern India
1750-60	Aoki Konyo, a Japanese scholar, compiles a Dutch-Japanese dictionary (1758)	Robert Clive defeats Siraj-ud-daula, Nawab of Bengal, at Battle of Plassey (1757)
1760-70		
1770-80		
1780-90	British export of opium* from India to China expands dramatically	
1790-1800		Ranjit Singh* founds Sikh kingdom in Punjab (1799)
1800-10		
1810-20		
1820-30	Javanese revolt against Dutch (1825-30)	Practice of *sati* made illegal (1829)
1830-40	Ottoman sultan Abdul Majid starts a programme of modernisation (1839)	
1840-50		
1850-60	King Rama IV rules Thailand, opens the country to foreign trade (1853)	Railway and telegraph line introduced (1853); the Great Revolt* (1857)
1860-70	French begin to occupy Indo-China (Southeast Asia) (1862)	
1870-80	Opening of the first Japanese railway, Tokyo to Yokohama (1872)	Famine in the Deccan, southern India (1876-78), over 5 million die
1880-90	Britain annexes Burma (Myanmar) (1885-86)	Foundation of Indian National Congress* (1885)
1890-1900		
		

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DATES	ASIA	SOUTH ASIA
1900-10	Japanese navy defeats Russian fleet (1905)	
1910-20	Balfour Declaration promises homeland for Jews in Palestine (1917)	
1920-30		Non-Cooperation Movement (1921) launched by Mahatma Gandhi; E. V. Ramaswamy Naicker launches the Self-Respect Movement in Tamil Nadu (1925)
1930-40	Opening of British oil pipeline from Iraq to Syria (1934)	Alam Ara by Ardeshir Irani (1931) is the first Indian talkie. Berlin–Baghdad Railway linking Baghdad to Istanbul begins operation (1940)
1940-50	USA drops atom bombs on Japanese cities of Hiroshima and Nagasaki* (1945) killing approximately 120,000 civilians. Many more were to die later through the effects of radiation; formation of People's Republic of China (1949)	Quit India Movement (1942); India and Pakistan become independent (1947)
1950-60	Bandung Conference (1955) strengthens the Non-Aligned Movement	India becomes a republic* (1950)
1960-70	Arab leaders set up Palestine Liberation Organisation to unite Palestinian refugees (1964); war in Vietnam (1965-73)	Sirimavo Bandarnaike* becomes world's first woman prime minister (1960) 
1970-80	Shah of Iran is overthrown (1979)	Bangladesh emerges as an independent nation (1971)
1980-90	Mass demonstrations for democracy in Tiananmen Square, Beijing, China (1989)	A leak at the Union Carbide pesticides plant in Bhopal (1984) leads to one of the worst industrial disasters in history, thousands die
1990-2000	Gulf War between Iraq, Kuwait and the USA	India and Pakistan conduct nuclear tests (1998)
		

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'Overcoming Modernity'

State-centred nationalism found full expression in the 1930s and 1940s as Japan launched wars to extend its empire in China and other parts of Asia, a war that merged into the Second World War after Japan attacked the USA at Pearl Harbor. This period saw greater controls on society, the repression and imprisonment of dissidents, as well as the formation of patriotic societies, many of them women's organisations, to support the war.

An influential symposium on 'Overcoming Modernity' in 1943 debated the dilemma facing Japan – of how to combat the West while being modern. A musician, Moroi Saburo, posed the question of how to rescue music from the art of sensory stimulation and restore it to an art of the spirit. He was not rejecting Western music but trying to find a way that went beyond merely rewriting or playing Japanese music on Western instruments. The philosopher Nishitani Keiji defined 'modern' as the unity of three streams of Western thought: the Renaissance, the Protestant Reformation, and the rise of natural sciences. He argued that Japan's 'moral energy' (a term taken from the German philosopher Ranke) had helped it to escape colonisation and it was its duty to establish a new world order, a Greater East Asia. For this a new vision that would integrate science and religion was necessary.

ACTIVITY 2

Would you agree with Nishitani's definition of 'modern'?

After Defeat: Re-emerging as a Global Economic Power

Japan's attempt to carve out a colonial empire ended with its defeat by the Allied forces. It has been argued that nuclear bombs were dropped on Hiroshima and Nagasaki to shorten the war. But others think the immense destruction and suffering it caused were unnecessary. Under the US-led Occupation (1945-47) Japan was demilitarised and a new constitution introduced. This had Article 9, the so-called 'no war clause' that renounces the use of war as an instrument of state policy. Agrarian reforms, the re-establishment of trade unions and an attempt to dismantle the *zaibatsu* or large monopoly houses that dominated the Japanese economy were also carried out. Political parties were revived and the first post-war elections held in 1946 where women voted for the first time.

The rapid rebuilding of the Japanese economy after its shattering defeat was called a post-war 'miracle'. But it was more than that – it was firmly rooted in its long history. The constitution was democratised only now, but the Japanese had a historic tradition of popular struggles and intellectual engagement with how to broaden political participation. The social cohesion of the pre-war years was strengthened, allowing for a close working of the government, bureaucracy and industry. US support, as well as the demand created by the Korean and the Vietnamese wars also helped the Japanese economy.

used force to expand its lucrative trade in opium leading to the first Opium War (1839-42). This undermined the ruling Qing dynasty and strengthened demands for reform and change.

The Opium War: A European painting.



THE OPIUM TRADE

The demand for Chinese goods such as tea, silk and porcelain created a serious balance-of-trade problem. Western goods did not find a market in China, so payment had to be in silver. The East India Company found a new option – opium, which grew in India. They sold the opium in China and gave the silver that they earned to company agents in Canton in return for letters of credit. The Company used the silver to buy tea, silk and porcelain to sell in Britain. This was the 'triangular trade' between Britain, India and China.

ACTIVITY 3

Does this painting give you a clear sense of the significance of the Opium War?

Qing reformers such as Kang Youwei and Liang Qichao realised the need to strengthen the system and initiated policies to build a modern administrative system, a new army and an educational system, and set up local assemblies to establish constitutional government. They saw the need to protect China from colonisation.

The negative example of colonised countries worked powerfully on Chinese thinkers. The partition of Poland in the eighteenth century was a much-discussed example. So much so that by the late 1890s it came to be used as a verb: 'to Poland us' (*bolan wo*). India was another such example. In 1903, the thinker Liang Qichao, who believed that only by making people aware that China was a nation would they be able to resist the West, wrote that India was 'a country that was destroyed by a non-country that is the East India Company'.

The Rise of the Communist Party of China

When the Japanese invaded China in 1937, the Guomindang retreated. The long and exhausting war weakened China. Prices rose 30 per cent per month between 1945 and 1949, and utterly destroyed the lives of ordinary people. Rural China faced two crises: one ecological, with soil exhaustion, deforestation and floods, and the second, a socio-economic one caused by exploitative land-tenure systems, indebtedness, primitive technology and poor communications.

The CCP had been founded in 1921, soon after the Russian Revolution. The Russian success exercised a powerful influence around the world and leaders such as Lenin and Trotsky went on to establish the Comintern or the Third International in March 1918 to help bring about a world government that would end exploitation. The Comintern and the Soviet Union supported communist parties around the world but they worked within the traditional Marxist understanding that revolution would be brought about by the working class in cities. Its initial appeal across national boundaries was immense but it soon became a tool for Soviet interests and was dissolved in 1943. Mao Zedong (1893-1976), who emerged as a major CCP leader, took a different path by basing his revolutionary programme on the peasantry. His success made the CCP a powerful political force that ultimately won against the Guomindang.

Mao Zedong's radical approach can be seen in Jiangxi, in the mountains, where they camped from 1928 to 1934, secure from Guomindang attacks. A strong peasants' council (soviet) was organised, united through confiscation and redistribution of land. Mao, unlike other leaders, stressed the need for an independent government and army. He had become aware of women's problems and supported the emergence of rural women's associations, promulgated a new marriage law that forbade arranged marriages, stopped purchase or sale of marriage contracts and simplified divorce.

In a survey in 1930 in Xunwu, Mao Zedong looked at everyday commodities such as salt and soya beans, at the relative strengths of local organisations, at petty traders and craftsmen, ironsmiths and prostitutes, and the strength of religious organisations to examine the different levels of exploitation. He gathered statistics of the number of peasants who had sold their children and found out what price they received – boys were sold for 100-200 yuan but there were no instances of the sale of girls because the need was for hard labour not sexual exploitation. It was on the basis of these studies that he advocated ways of solving social problems.

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MAP 2: The Long March

The Guomindang blockade of the Communists' Soviet forced the party to seek another base. This led them to go on what came to be called the Long March (1934-35), 6,000 gruelling and difficult miles to Shanxi. Here, in their new base in Yanan, they further developed their programme to end warlordism, carry out land reforms and fight foreign imperialism. This won them a strong social base. In the difficult years of the war, the Communists and the Guomindang worked together, but after the end of the war the Communists established themselves in power and the Guomindang was defeated.



Photograph of soldiers on the Long March reclaiming wasteland, 1941.

Establishing the New Democracy: 1949-65

The Peoples Republic of China government was established in 1949. It was based on the principles of the 'New Democracy', an alliance of all social classes, unlike the 'dictatorship of the proletariat'* that the Soviet Union said it had established. Critical areas of the economy were put under government control, and private enterprise and private ownership of land were gradually ended. This programme lasted till 1953 when the government declared that it would launch a programme of socialist transformation. The Great Leap Forward movement launched in 1958 was a policy to galvanise the country to industrialise rapidly. People were encouraged to set up steel furnaces in their backyards. In the rural areas, people's communes (where land would be collectively owned

*This term was used by Karl Marx to stress that the working class would replace the repressive government of the propertied class with a revolutionary government and not a dictatorship in the current sense.

Full diplomatic relations and embassies are not possible as Taiwan is considered to be part of China.

The question of re-unification with the mainland remains a contentious issue but "Cross Strait" relations (that is between Taiwan and China) have been improving and Taiwanese trade and investments in the mainland are massive and travel has also become easier. China may be willing to tolerate a semi-autonomous Taiwan as long as it gives up any move to seek independence.

The Story of Korea

Beginnings of Modernisation

During the late nineteenth century, Korea's Joseon Dynasty (1392–1910) faced internal political and social strife and increasing foreign pressure from China, Japan and the West. Amidst this, Korea implemented modernisation reforms in its governmental structures, diplomatic relations, infrastructure and society. After decades of political interference, the imperial Japan annexed Korea as its colony in 1910, bringing the over 500-year long Joseon Dynasty to its end. However, the Korean people were angry about Japan's suppression of their culture and forced assimilation. Desiring independence, Koreans around the country demonstrated against the colonial rule, set up a provisional government and sent delegations to appeal to foreign leaders at international meetings, such as the Cairo, Yalta and Potsdam conferences.



Koreans celebrate their independence from Japan in 1945.

The Japanese colonial rule ended after 35 years in August 1945 with Japan's defeat in the World War II. However, it was the continued efforts of independence activists both inside and outside Korea that ensured Korea's independence after Japan's defeat. Following liberation, the Korean Peninsula was temporarily divided along the 38th parallel with the Soviets managing the North and the U.N. managing the South even as the nations worked to disband the Japanese forces in the region. However, this division became permanent as separate governments were established in both the North and the South in 1948.

A Post-War Nation

In June 1950, the Korean War broke out. With South Korea receiving support from the US-led United Nations forces and North Korea receiving support from communist China, it developed into

a vintage proxy war of the Cold War era. In July 1953, after three years, the war ended in an armistice agreement. Korea remained divided. The Korean War had caused not only massive losses of life and property, but also a delay in free-market economic development and democratisation. Prices suddenly rose due to inflation caused by increased national expenses and currency issued during the war. Furthermore, industrial facilities constructed during the colonial period had been destroyed entirely. As a result, South Korea was forced to rely on the economic assistance being provided by the USA.

Though South Korea's first president Syngman Rhee had been elected in 1948 through democratic process after the Korean War, he extended his administration, twice through illegal constitutional amendments. In April 1960, citizens protested against a rigged election in what is known as the April Revolution and Rhee was forced to resign.

With the revolution as an impetus, the spirit of the people, which had been suppressed during the Rhee administration, erupted in the form of demonstrations and demands. However, the Democratic Party administration, which took power after Rhee's resignation could not properly respond to citizens' demands due to internal divisions and conflict. Rather, reformist political powers emerged and the students' movement grew into a unification movement. This was not looked upon favourably by the military authorities. In May 1961, the Democratic Party government was overthrown in a military coup staged by General Park Chung-hee and other military authorities.

Rapid Industrialisation under Strong Leadership

In October 1963, an election was held and military coup leader Park Chung-hee was elected the president. The Park administration adopted a state-led, export-oriented policy to achieve economic growth. The five-year economic plans of the government favoured large corporate firms, placed emphasis on expanding employment and increased Korea's competitiveness.

Korea's unprecedented rate of economic growth began in the early 1960s when the state policy shifted from import substitution industrialisation (ISI) towards a focus on exports. Under the export-oriented policy, the government supported labour-intensive light industrial products, such as textiles and garments in which Korea had a comparative advantage. During the late 1960s and 1970s, the focus again shifted from light industries to value-added heavy and chemical industries. Steel, non-ferrous metals machinery, shipbuilding, electronics and chemical production were selected as the most important industries in the race for economic growth.

environmental conditions which influence the language and culture of people living in these regions. Thus, India houses maximum number of written and orally alive languages in the world. The diversity of languages, cultures and traditions makes India one of the most tolerant and harmonious country.

India is home to five major language families. These language families are: Indo-Aryan, Dravidian, Austro-Asiatic, Tibeto-Burmese and Semito-Hamitic. India's rich linguistic and cultural heritage is mirrored in her literatures, both oral and written.

Sanskrit belongs to Indo-European group of languages. Sanskrit was gradually standardised and given a highly scientific grammar by Panini, the great grammarian, in about fifth century B.C. Sanskrit was the language of religion, philosophy and learning. The people spoke a number of dialects which are called *Prakrits*. Buddha preached in the language of the people. Buddhist literature was written in *Pali*, one of the *Prakrits*. Among the Dravidian languages, Tamil is the most ancient. The others developed during the first millennium of the Christian era. Though Sanskrit again became the predominant language of learning in the period of the Guptas, the *Prakrits* continued to develop. The various spoken languages that developed are called *Apabhramshas*. These formed the basis of the modern Indian languages which developed in various regions of India during the medieval period.

Literature and Cultural Expression

Literature presents the panorama of the history of mankind; the social institutions, religious beliefs, scientific achievements and philosophical ideas. The objective is to present the journey of thoughts from the earliest times to the present times. The literature in a variety of forms appeals to the readers because it relates to the questions of human life. There is a unique linguistic variety in literature but it keeps pace with the cultural evolutions in the society and the common thread runs across the literatures creating a continuum of the progress of human civilisation. It was the advent of British era that marked the change in the course of language and literature. The tradition of nurturing plurality of languages strengthened in the modern era, enriching almost all the regional languages with a rich oral and written body of literature. The establishment of printing press by 1800 brought remarkable changes in the field of communication. The author could communicate directly with the reader in the languages of the people. The rise of journalism with the introduction of newspapers and magazines helped the development of prose writing which was so far a neglected



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field. Earlier the British felt that for expansion of their power learning local Indian languages would be of great benefit but by 1835, Macaulay's Minute on Education changed their thinking and English began making its presence felt in the political and social life of the people of India by the end of 19th century English. The contact with West opened new avenues in the realm of socio-political thought, themes and forms such as free verse. In the recent years, more Publications in English by Indian writers are being seen on bookshelves. Many Indians started writing in English as well. Henry Derozio and Michael Madhusudan Dutt are the pioneers in this field.

There is however, no complete departure from the tradition, in that many modern writers, including those who wrote in English, continue to draw their inspiration and themes from the classical epics and other texts. Several Indian writers have written par excellence not only in traditional Indian languages but also in English. India's Nobel laureate in literature was the Bengali writer Rabindranath Tagore, who wrote some of his work originally in English, and did some of his own English translations from Bengali. Writers like Vikram Seth, Raja Rao, Anita Desai, Shashi Deshpande, R.K. Narayan, Ruskin Bond, derived inspiration from Indian themes.

Here is the translation of a Tamil poem in English. Translation, as a creative genre, is the characteristic of modern literature.

Poetry: Kuruntokai Poem 312
Poet: Kapilar

What He Said

My love is a two-faced thief.
In the dead of night
she comes like the fragrance
of the Red-Speared Chieftain's
forest hills,
to be one with me.

And then, she sheds the petals
of night's several flowers,
and does her hair again
with new perfumes and oils,
to be one with her family at dawn
with a stranger's different face.

Translated by A.K. Ramanujan

Share your experiences with peers about reading literature. You can explore on the basis of the following:

- Reading literature gives pleasure when it connects with our lives and surroundings
- Reading about something unfamiliar is exciting and thought provoking
- The focus in reading is on the thought and ideas, not how the sentences are constructed
- Discuss at least two works of an author of your choice

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We may tabulate the history of Indian literature as:

- *Vedic* Literature, approx. up to 1200 B.C.E.;
- Classical Literature, from 1200 B.C.E. to fifth C.E. (in classical Sanskrit, *Pāli*, *Prākṛt* and Tamil);
- *Prākṛt* Literature, from 1st century C.E. to eleventh century (in various *Prakrits*);
- *Apabhramsa* Literature, from seventh century CE to eighteenth century (regional Indian languages' literatures);
- Indian-language literatures in modern period, from eighteenth century

A Mosaic of Language and Literature in India

The history of Indian literature is ancient and vast. It had been an instrument of instruction since antiquity. *Śruti* and *Smṛti* literature, *Sūtra literature, Jātaka tales, Pancatantra, Kathasaritsāgara, Thirukural, Athichudi and Vachanas* are examples of traditions of literature which have enriched human life encouraging them to follow human values, and living in harmony with nature. In India, in the sixth and seventh century, A.D., there were great literary skills in epic poetry and drama. Indian intellectuals explored the field of medicine, astronomy, geometry, law and many others. The Indian thinkers explored religion and philosophy with vigour and originality in the interest of the people.

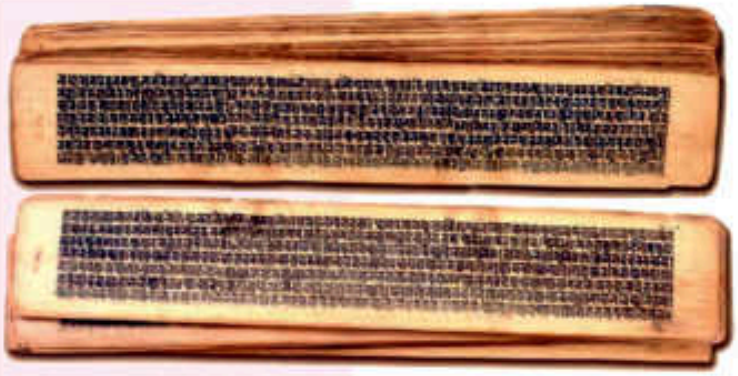
Ancient period falls roughly between 2000 B.C. to A.D. 1000. Sanskrit, Tamil, Pali and Prakrit were the main languages used for writing in ancient times. Other languages of antiquity include Kannada, Ardhamagadhi and Apabhramsa. Sanskrit literature from ancient period can be divided into the following.

Vedic Age

From this pristine antiquity, we receive two types of literatures namely 'Śruti' (heard and revealed) literature and 'Smṛti' (remembered and later recorded) literature. The four *Vedas—Ṛgveda, Yajurveda, Atharvaveda* and *Sāmveda* along with eleven *Brāhmaṇas*, three *Araṇyakas* and more than 100 *Upaniṣads* belong to the *Shruti* form. *Smṛti* texts are ascribed to human authorship and contain six *Vedāṅgas* (adjuncts to the *Vedas*), the epics—the *Rāmāyaṇa*, the *Mahābhārata* and the *Purāṇas*. The *Vedas* describe procedures for *yajñas*, sacrifice and prescribe rituals and mantras to invoke powers of various gods to attain success in human efforts. All human effort in *vedic* way of life is put into four *puruṣārthas* ('purpose of life on earth')—*Dharma, Artha, Kāma* and *Mokṣa*. *Upaniṣads* are contemplative enquiries into the mysteries of



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Rgveda (padapāṭha) manuscript in Devanagari, early 19th century

Source: *Vedparijat*, (August, 2014), NCERT









life. The two epics are narratives on the legendary life of Rāma and Kṛṣṇa—the famous avatāras of Lord Viṣṇu.

Post Vedic age

In the post Vedic period, Sanskrit literature evolved in the form of drama, verse and prose. Significant playwrights of the period were Bhāsa, Kālidāsa, Śūdraka and Bhavabhuti. Bhāsa's plays largely adopted their subject from the epics of the *Rāmāyaṇa* and the *Mahābhārata*. Thirteen of his plays are extant and some of them are: *Svapanavāsavadattam, Cārudattam, Abhiṣekanāṭakam, Pratimānāṭakam, Kaṇabhāram* and *Madhyamavyāyogam*. Kālidasa is considered as the greatest Sanskrit poet and playwright. His poem *Meghadūtam* and his play, *Abhijñānaśākuntalam* are world renowned. Here are some classic expressions from ancient Sanskrit literatures.

From the *Ṛgveda*

Devī Sūktam

सङ्गच्छध्वं संवदध्वं सं वो मनांसि जानताम्।
देवा भागं यथा पूर्वे सञ्जानाना उपासते।।

Saṅgacchadhvaṁ saṁvadadhvaṁ saṁ vo manāṁsi jānatām
devā bhāgaṁ yathā pūrve sañjānānā upāsate

May you move in harmony, speak in one voice, let your minds be in agreement just as the ancient gods shared their portions of sacrifice.

From Taittirīya Upaniṣad

वेदम् अनूच्य आचार्यः अन्तेवासिनम् अनुशास्ति
सत्यं वद, धर्मं चर, स्वाध्यायात् मा प्रमदः,आचार्याय
प्रियं धनम् आहृत्य प्रजातन्तुं मा व्यवच्छेत्सीः, सत्यात् न
प्रमदितव्यं, धर्मात् न प्रमिदतव्यम्, कुशलात् न प्रमदितव्यम्
भूत्यै न प्रमिदतव्यम्, स्वाध्यायात् न प्रमदितव्यम्।

vedam anūcya ācāryaḥ antevāsinam anuśāsti—satyam vada, dharmaṁ cara, svādhyāyāt mā pramadaḥ, ācāryāya priyaṁ dhanaṁ āhṛtya prajātantuṁ mā vyavacchetsīḥ, satyāt na pramaditavyaṁ, dharmāt na pramaditavyaṁ, kuśalāt na pramaditavayaṁ, bhūtyai na pramaditavyaṁ, svādhyāyat na prmaditavyam.

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This is the advice of teacher to his pupils at the end of their stay in Gurukul. The Guru urges them to follow the path of truthfulness and work for the welfare of others.

From Raghuvaṁśam by Kalidas

शैशवेऽभ्यस्तविद्यानां यौवने विषयैषिणाम्।
वार्धके मुनिवृत्तीनां योगेनान्ते तनुत्यजाम्।।

śaiśavebhyastavidyānāṁ yauvane viṣayaiṣiṇāṁ
vārdhake munivṛttīnāṁ yogenānte tanutyajām

- What are the features of Vedic and Post Vedic literature?
- What were the languages of literary writings during Vedic and Post Vedic Literature?
- Recite Devi Suktam with intonation.

Prākṛt, Pālī and *Apabhraṁsa*

'Prākṛt' ('natural' or 'common') was an important colloquial language which was used in Sanskrit drama for dialogues of women and minor characters. *Gāthā Sattasai* is a third century work in Prākṛt. Ashoka's inscriptions have also used Prākṛt along with Pali. Buddhist philosophical work like *Dhammapada* and educational stories like Jātaka Tales are in Pali. Lord Mahavira Jain used *Apabhraṁsa* to disseminate his teachings.

Ancient Tamil Literature

Ancient Tamil literature is known as Saṅgama Literature. The earliest known work in the ancient Tamil is *Agastyam*. Another seminal work Tholkapiayam (300 B.C.) is a treatise on Tamil grammar. 'Akam' ('interior landscape') and 'Puram' ('exterior landscape') forms of poetry also come from Sangam period. It conceptualised five landscapes, known as 'tinai' which denotes a literary category named after a flower native to that natural setting. The tinais are:

- *Kurinchi* (mountain and mountain related landscape),
- *Mullai* (forest and forest related landscapes),
- *Marutam* (cultivable field and field related landscapes), *Neytal* (sea and Sea related landscapes), *Palai* (dry, a kind of arid desert and desert related landscapes).

The two epics found in Tamil are *Cilapathikaram* and *Manimekalai*.

Regarded as one of the great works of Tamil literature, the *Silappadikaram* is a poetic rendition with details of Tamil culture; its varied religions; its town plans and city types; the mingling of different people; and the arts of dance and music.

Manimekalai is composed by the poet Chithalai Chathanar. *Manimekalai* is a poem in 30 cantos. It is a story of Kovalan and Mādhavī who became Buddhist Bhikkhuṇi.






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Ancient Kannada literature

Earliest work extant in Kannada literature is *Kavirājamārgam* from ninth century. Kannada language is known to exist since the third century B.C. Pampa, Sri Ponna and Ranna are three major Jaina poets who wrote in classical Kannada in the tenth century A.D. Ādikavi Pampa's *Rāmāyaṇa* in Kananda is the most significant source text for the poets of the succeeding generations.

- Describe the features of ancient Tamil literature.
- Name the three poets who influenced the Kannaḍa poetry.

The Epics

The two epics of India *Mahābhārata* and *Rāmāyaṇa* are the representations of cultural, religious and political lives of the people of Indian subcontinent. The epics are written in the literary tradition of *Kāvya*. The epics are the reflection of heroic age embedded with value and moral teachings for human beings. These epics have been an enduring part of visual and oral traditions of transmitting knowledge in India hence have a profound influence on the lives of people. The epics have been translated into many languages and dialects across the world.



Kannagi from the Tamil Epic Silapathikaram

The *Rāmāyaṇa* is considered as the foremost literature in classical Sanskrit and its author Vālmīki is known as the 'Ādikavi'. The *Rāmāyaṇa* is a saga of life of Lord Rama which is composed in 24000 verses. The epic is the source of inspiration for many poets in ancient, medieval and even the modern times. The *Mahābhārata* is written by *Vedavyāsa* who enriched our literature with significant literary and *shastric* treatises like the *Purāṇas* and *Brahmasūtras*, and perhaps the first to write *Vedas*. The *Mahābhārata* consists of one lakh verses which are supposed to be the largest work of literature in any time and in any language. Indian tradition treasures a legend saying that any knowledge related to humankind has a place in the *Mahābhārata*. The world famous *Bhagavad-Gīta*, consisting of 700 verses forms a part of the *Mahābhārata* in the 6th *Parva* (chapter) out of its 18 *parvas*. *Bhagavad-Gīta* is a complete philosophical doctrine which answers our personal emotional dilemmas. It shows threefold paths to seeking and attainment of salvation (*mokṣa*) which are the paths of *Karma*, *Jñāna* and *Bhakti*.

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The Rāmāyaṇa became the symbol of life of prosperity, *dharma*, harmony and spiritualty—*rāmarajya* for the people representing Indian philosophy and culture. There are many versions of the *Rāmāyaṇa* which differ in language but the core philosophy remains the same, touching the heart and soul of people. The *Rāmāyaṇa* also served as a source for themes for later writers. The great writer Kalidasa drew inspiration for his literary works like *Kumārasambhavam* and *Meghdūtam* from the *Rāmāyaṇa*.

Classical Languages of India

A language is accepted as a classical language, if it has an independent tradition, antiquity and rich repertoire of classical literature. A classical language should be ancient and it should have its own independent tradition, not influenced by any other tradition. It must have a large and extremely rich body of ancient literature. The Sahitya Academy of India prescribed the following four criteria for a classical language in the year 2004.

- The high antiquity of early textual history of over 1500 to 2000 years.
- A body of ancient literature that is considered a valuable heritage by generations of speakers.
- The literary tradition should be original and not borrowed from another speech community.
- The classical language and literature should be distinct from the modern and there may also be a discontinuity between the classical language and its later forms of offshoots.

Government of India accorded classical status to some languages in India based on their antiquity and their traditions. Languages have been given the status of classical language. Given in bracket are the years in which the language was given the status—Tamil (2004), Sanskrit (2005), Telugu (2008), Kannada (2008), Malayalam (2013) and Odiya (2014).






The Rāmāyaṇa

As the story goes, the aged king Daśaratha chooses one of his four sons, Rāma, to succeed him as the ruler of Ayodhyā, but is finally persuaded by one of his wives to name Rāma's brother Bharata as his successor. Bowing to his father's decision, Rāma prepares to leave the capital and his wife. Sītā refuses to accept this separation and insists on accompanying Rāma and his brother Lakṣmaṇa into exile.

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The city was distressed and citizens were dazed
And crowds unending followed as he went.
Rāma arrived where Jānaki was.
Startled, she rose, for she did not understand
Why the crowd did weep or why they came
Thus dazed and covered with dust.
Nor why her prince was strangely clad.
Her fright increased
As the women about embraced her;
They would not speak but smothered her in tears.
She turned her large eyes then to Rāma:
"My prince, tell me now,
Is the king well
Or has some illness stricken him?"
And Rāma quietly spoke:
"My peerless brother beloved will rule the realm
By the command of them that gave me being.
I go today to see the hills
From where the rains are sent.
And till I come again abstain from grief."
She sobbed but not for his dread exile
Or for the kingdom's sake:
She sobbed for the cruel words that scalded her ears:
"Grieve not the time I shall be away!"
For how could Sita bear the thought
Of separation? Were they not
An eternal pair together in the sea of milk
Together again on earth when he chose to be born...

Ayodhyā Canto of the *Rāmāyaṇa* by Kamban, one of the greatest classic Tamil poets of the ninth century. Translated from Tamil by Sri Chakravarti Rajagopalachari.

Fables in Ancient Literature

Pañcatantra

The *Pañcatantra* is a collection of ancient India fables. It is believed that Vishnu Sharma composed them in first century C.E. It has influenced the composition of literature in other parts of the world, for example the writing of Aesop Fables. *Pañcatantra* fables are marked by a moral. The idea that an action has consequences is well portrayed in these fables. The fables are widely popular even today because of their simplicity, logic and human desire to follow morals and values in life.

Here is a sample of its style

A donkey was employed in pulling a washerman's cart and after the day's labour, at nightfall, he liked to have a good feed off the neighbour's cucumber field. A jackal once joined him,

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and when the two had feasted on the cool and delicious fruit, the donkey exclaimed: "Isn't it a glorious night, old fellow? I feel so jolly, I must sing a song." The wise jackal thought that they should leave the field stealthily keeping quiet, but the silly ass brayed merrily, until the farmer then woke up and gave him a sound thrashing.

Jātaka Tales



Jātaka Tales

Source: https://search.creativecommons.org/photos/fdf30c26-6fee-467b-8fc5-fce66cf89229

Jātaka stories emphasise Buddha's great abilities as a visionary and storyteller. The stories are marked by moral lessons. The stories are important to know the culture of the people, and also inspire literature, theatre opera and other art forms. Jātakas are the birth stories of Buddha. Jātaka tales are said to be one of the most authentic account of life in ancient India. The stories capture the various incarnations of Buddha, practical wisdom, lives of people in frames of hope, happiness and despair. The stories present life in bazaar, caravan, field barracks and life lived in quiet and thought mediations.

Here is a sample of its style

Prince Suvaṇṇa Sāma (the devoted prince) — Act of benevolence

Sāma's parents lost their sight when he was 16 years old. From then on he was their only help. He took care of his parents. Every day he went to the Migasammata River to gather fruit and fetch water in a pot. At that time King Piliyakkha of Benares in his great desire for venison, also



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came to the river and at last reached the spot where Sāma stood. Seeing Sāma taming the wild animals, the King was wondering whether the creature was a god or a Nāga? To find out, he decided to wound and disable him, and then ask. The King shot a poisoned arrow and wounded Sāma, who fell to the ground. Sāma spoke: "I have no enmity against anyone. Who has wounded me?" The King went to Sāma and asked him his name. Sāma then told the King the story of fostering his blind parents. The King thought to himself, "I have done evil to such a holy being; how can I comfort him?" He decided to bring the blind parents to their son. In the meantime the goddess Bahusodarī decided to help Sāma and his parents. In the end Sāma recovered from his wound and both of his parents' sight was restored.

Hitopadeśa

Hitopadeśa is a book of worldly wisdom presented through the characters of birds, animals and humans. The purpose of the book appears to encourage proficiency in Sanskrit expression and the knowledge of wise behaviour (*nīti-vidya*). The stories are elaborate, ending with moral messages. However, very peculiar to its style, it has condensed and crisp verses in between the stories to illustrate the points made by the characters.

Hitopadeśa is structured in four books with a preface (*prāstāvika*). The four books are as follows:

- *Mitralābha*
- *Suhṛdbheda*
- *Vigraha*
- *Sandhi*

Here is a sample of a story from the book *Suhṛdbheda*

The Story of the Black Snake and the Golden Chain

"A pair of crows had their abode in a certain tree, the hollow of which was occupied by a black snake, who had often devoured their young. The hen-bird, finding herself breeding again, thus addressed her mate: 'Husband, we must leave this tree; we shall never rear young ones while this black snake lives here! You know they say—

'From false friends that breed thee strife,
From a house with serpents rife,
Saucy slaves and brawling wife—
Get thee out, to save thy life.'

'My dear,' replied the crow, 'you need not fear; I have put up with him till I am tired. Now I will put an end to him.'

'How can you fight with a great black snake like that?' said the hen-bird.

'The Lion proud died by the hare so meek.'

'How came that about?' asked the hen-crow.

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Doubt nothing,' answered the other— 'He that hath sense hath strength; the fool is weak…'

The Book of Good Counsels: From the Sanskrit of the Hitopadesa, by Sir Edwin Arnold, M.A. 1861

- Read and describe Ayodhyā Canto of the *Rāmāyaṇa* by Kamban in your words.
- What is the common thread in the fables of *pañcatantra*, *jātaka* and *Hitopadeśa*?

Medieval Period

Emergence of the *Bhakti* Movement

In the A.D. sixth century, *Bhakti* emerged in the Tamil land and there was an upsurge of *bhakti* poetry. Compositions of *Alvars* (the word literally means immersed in God) and *Nayanmars* are expressions of *Vishnu* and *Shiva bhakti*. In Kannada region *Basavanna* (1105–68) in the twelfth century began the movement in response to the prevailing caste hierarchy. The bhakti literature is known as *Vacanasāhitya*. *Akkamahadevī*, *Allama Prabhuare* the proponents of *Bhakti* movement. Jñānadeva, Nāmadeva and Tukārām contributed profusely to this movement connecting to people in their language in Marathi.

In northern India Kabir emerged as a great force in lending support to Bhakti movement. His verses were like a mirror to the hierarchy ridden society of that time. He spoke with all humility and faith in God. Another believer in *Bhakti* was Guru Nanak who tried to synthesise the different religious faiths to serve as a guide for humanity. He preached devotion to God than rituals.

Poet-saints who produced literary masterpieces include Chaitanyadeva in Bengali, Surdas in Hindi, Mira Bai in Rajasthani, Jnaneshwar in Marathi and Narsinha Mehta in Gujarati. The contribution of women writers needs special mention. The Kashmiri compositions of Lal Ded and Habba Khatoon are known for their mystical flavour and intense agony of expression, Mirabai's compositions in Rajasthani is known for a unique devotional quality, Andal's mysticism in Tamil and Akka Mahadevi's vachanas in Kannnada are testimonies to the great contribution of women to literature of the time. Besides this, medieval times were characterised by religious and cultural synthesis that integrated the Islamic and Sufi elements into the culture and literature of Northern and Western India.



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Renderings of Bhakti Poets

Guru Nanak

Guru Nanak was a spiritual Guru whose teachings were expressed in devotional hymns. He was a *Nirguna Bhakti* saint and a social reformer. He was an opponent of caste and religious distinctions. Guru Nanak firmly believed and preached that the most important form of worship is *bhakti*—loving devotion to God. He was the first Sikh Guru and the founder of Sikhism.

Gaganmahthāl, rav-cāṅddīpakbaney
tārkamaṅdaljanakmotī
dhūpmaliāṅlopavancāvarokarey
sagalbanraiphūlantjyoti.

Translation—The Sky is your platter,
The sun and moon are the dīpakas [lamps or lights],
The stars in the sky are the pearls,
The dhūp [incense] is the fragrance
That the wind propels,
The whole forest is your flowers.

Kabir

Kabir was a mystic poet and saint. He represented *Nirgun* stream of Bhakti movement. Kabir is revered for his verses which expressed his philosophy with simplicity and honesty. There is no evidence that he received any formal education. Kabir wrote from his experiences which he gained through his travels and *satsangs*. He opposed distinctions of caste and creed and propagated harmony among people.

Aisī vāṇī boliye, mana ka āpā khoye
Apnā tan śītal kare, auran ko sukh hoye.

Translation—Speak such words, sans ego's ploy
Body remains composed, giving the listener joy.

Lal Ded

Lalleshwari, locally known mostly as Lal Ded, was a Kashmiri mystic of the Kashmir Shaivism school of Philosophy in the Indian subcontinent. She was the creator of the style of mystic poetry called vatsun or Vakhs, literally 'speech'.

Another feature of this period is the emergence of Urdu as a language of confluence of Persian and Hindustani that marked the beginning of poetry forms like *qasida*, *marsiya* and *masnavi*, which found ready acceptance in Urdu and Kashmiri. It is interesting to note that when the Christian Missionaries came to India and wanted to compose on biblical themes, some of them adopted indigenous forms like *puran* and *pana*.

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Performing Art Traditions in India

3

Arts have always been very important in human life. Basic need of human beings to express gets communicated to others through various arts. Recent researches throw light on how during the prehistoric period people used to express through art forms. Art is a diverse range of human activities resulting out of creative skills and imagination. *Vātsyāyana* has described sixty four arts. Interestingly, in that list, the first four are performing arts—vocal music, instrumental music, dance and theatre. This also indicates that a strong tradition of Performing Arts can be traced back to the old civilisation of India. In broader sense, performing arts require skills in which the artistic expression is conveyed to the audience through performer's voice, body gestures or through sound objects or musical instruments. In everyday life, it is done through folk music, folk dance, folk theatre such as *Jātrā, Nauṭaṁki,* etc., by the common people. To practice the art in a specific specialised manner, it is referred to as the classical art forms such as, music, dance and theatre which has evolved over the centuries having a strong tradition with laid down rules and regulations.

Do you know that classical music has evolved from folk music. This signifies the strong connection between the two.

Performing arts occupy a significant place in the Indian society. There can be no doubt about the fact that art forms, be it performing or visual, reflect the thinking process of



Source: *Pragetihasik Bhartiya Chitrakalaa Mein Sangeet*

the society which comprises its people, their habitat, ethos, emotions, and uniqueness of communities and surroundings.



Source: *Pragetihasik Bhartiya Chitrakalaa Mein Sangeet*









Music in India

Indian music, i.e., *Bhāratīya Saṅgīta* has built a strong background over different occasions simultaneously enriching our culture. It has a rich treasure of classical, regional and folk forms expressed through vocal music and instrumental music. Music comprises three art forms, i.e., *Gīta, Vādya* and *Nṛtya* as referred in *Saṅgīta Ratnākara* written by Pt. Sharangdev "गीतं, वाद्यं त्रयं संगीतमुच्यते" In *Saṅgīta Pārijāta*;, Pt. Ahobal has said: "गीतावादित्रनृत्यानां रक्ति साधारण गुणः अतो रिक्तविहीनं यत्न तत् संगीतमुच्यते ".

Indian Music has been continuously affected by the social and religious conditions. It has gradually evolved through different periods and stages. The development of Indian music can be classified into three main periods:

1. Ancient Period
2. Medieval Period, and
3. Modern Period

Ancient Period (2500 B.C.–A.D. 1200)

The origin of Indian music can be traced back to evidences from the Indus Valley Civilisation. This period is also known as Vedic period. During this period, the Vedic hymns were chanted and some of them were also set to tune and rhythm. The rhythmic recitations of *Ṛgveda* were known as *Ṛcās* (ऋचायें). *Sāmaveda* is the compilation of these selected *Ṛcās* set to *Svaras* keeping their proposed *Chanda* (छंद) or rhythmic meters. Only three *Svaras* — *Udātta* (उदात्त), *Anudātta* (अनुदात्त) and *Svarita* (स्वरित) were used in *Sāmagāna*. *Udātta* was the sharp pitch, *Anudātta* was grave pitch and *Svarita* combined in itself the characteristics of both the pitches.

In Pāṇinīya Śikṣā, Pāṇini has mentioned two additional *Svaras* — 1. Uccaistara, higher than the *Udātta* and 2. *Sannatara* lower than the *Anudātta*. Further, seven notes evolved from these three Vedic *Svaras*. According to *Pāṇinīya Śikṣā*:

"उदात्ते निषाद-गांधारौ, अनुदात्ते ऋषभ-धैवतौ
शेषास्तु स्वरिताः गेयाः षड्ज-मध्यम-पंचमाः"

which means *Niṣāda (Ni,* the seventh *svara)* and *Gāndhāra (ga,* the third *svara)* arise from *Udātta; Ṛṣabha (Re,* the second *svara)* and *Dhaivata* (*Dha,* the sixth *svara*) arise from *Anudātta;* and *Ṣadja (Sa,* the first *svara*), *Madhyama (Ma,* the fourth *svara)* and *Pañcama (Pa,* the fifth *svara)* originate from *Svarita*.



Dancing Shiva

Source: https://nroer.gov.in/

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- *Mantras* chanted during religious activities with *Svaras* were known as *Sāmagāna* (सामगान).
- Seven *Svaras* (notes) evolved from three basic *Svaras* of *Sāmagāna*.
- *Sāmagāna* was sung in three different pitches with a variety of intonation used during recitations by a group of singers known as *Sāma Gāyakas*.
- Musical Instruments like *Ghoṣa, Vīṇā, Kaśyapī, Audumbari, Veṇu, Dundubhi, Puśkara, Tunāva, Āḍambara*, etc., were in practice.

There are two streams of music known as *Mārgī* and *Deśī*.

1. *Mārgī* or *Gāndharva Saṅgīta* was practiced for salvation.
2. *Deśī Saṅgīta* which was further divided into classical, semi-classical, folk music, etc.

Gradually *Sāmagāna* developed into *Gāndharva* which was further developed during the *Rāmāyaṇa*, the *Mahābhārata* and *Purāṇas*.

The epic *Rāmāyaṇa* composed by *Vālmīki* has the reference of terms like *Saṅgīta, Svara, Laya, Tāla, Mātrā, Mūrchanā, Jāti, Mārg Saṅgīta* and *Gāndharva*. The reference of musical instruments is also available in the epic, such as, *Vipañci, Vallakī*, etc. *Mahābhārata* written by *Kṛṣṇadvaipāyana Vyāsa*, has musical terms such as *Grāma, Mūrchanā* and the names of seven basic *Svaras* such as *Ṣaḍja, Ṛṣabha*, etc. It also has the reference of musical instruments used during this period, such as *Bherī, Jharjharī, Tūrya, Vīṇā*, etc.

1. Let us find a few *ślokas* of the Vedic period along with their meaning which we recite even today.
2. What are the musical evidences available in Vedic period and the epics— *Rāmāyaṇa* and *Mahābhārata*.

Bharat's Nāṭyaśāstra is the most important and pioneer work of music, dance and drama. It has 36 chapters, out of which six chapters (28th–33rd) are related to music. *Śruti, Grāma, Grāmarāgas, Jāti, Mūrchanā, Gīti, Alaṅkāra*, etc., have been discussed in depth in the book. *Dhruva Gīta* had a special place in *Nāṭya*. These were used in between different scenes called *Nāṭya Sandhis*. Concept of *Rasa* for aesthetic experience has also been discussed elaborately in *Nāṭyaśāstra*. This is the first text available on the classification of Indian Instruments.



Raag Mala Chitra: Bhairvi
Source: Centre for Cultural Resources and Training (CCRT)









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Loka Vādya

Classification of Indian instruments

1. *Tat* (string instruments)
2. *Suśirā* (wind instruments)
3. *Avanādha* (percussion instruments)
4. *Ghana* (instruments made of brass or wood)

In Matanga's *Bṛhaddeśī* (seventh to eighth centuries), we get to see the description of *Deśī Rāga* for the first time.

Matanga was the first to place frets (सारिका) on the *Kinnarī Vīnā*. The concept of Time Theory of *Rāgas* was first mentioned by Narada in *Saṅgīta Makaranda* (eighth to ninth centuries). The other important Treatises of ancient periods are *Naradī ya Śikṣā, Saṅgīta Makranda, Dattilam, Gīta Govinda, Saṅgīta Samayasāra and Saṅgīta Pārijāta.*

1. Did you notice the evolution of music from the Vedic period to second century? Mention some of the noticeable changes.
2. Name three Vedic *Svaras* which were used in *Samagana.*
3. How many basic notes (*Svaras*) are there in Indian music?
4. Who is the writer of *Nāṭyaśāstra*? How many chapters does it contain?

Guru-Śiṣya Parampará or the Oral Tradition

Music in India has been passed on in a tradition described as *Guru-Śiṣya Parampará* (teacher-disciple tradition). In the *gurukula* system of education (from the Vedic Era to the medieval era), a pupil or *śiṣya*, after his initiation (sacred thread ceremony), lived in the house of his guru, or teacher, and studied the *Vedas* and other subjects under his guidance, for a period of 12 years. Gurus were expected to teach everything they knew to the disciple. The *Gurukula* was the precedent of the concept of *gharāna* in Hindustani music from the eighteenth to twentieth century, the difference being, in a *gharāna,* the learning was strictly in music and dance in a particular style or *śailī*. Knowledge in dance and music was passed on from one generation to the other through this oral tradition, there was no written word or process of documentation in earlier times. It evolved through successive generations. Even in the present times, music and dance classical traditions are learnt in the *guru-śiṣya parampará.*

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Medieval Period (A.D. 1201 – A.D. 1800)

The medieval period is known for the development of musical forms, musical instruments and documentation of music in a vast number of authentic texts which are available in the shape of *Grantha* (ग्रन्थ) to understand the growth of classical music.

Śāraṅgadeva (A.D. 1210–1247) was the author of the important musicological text *Saṁgīta Ratnākara*. In this text, the author describes *Saṁgīta* (music) as a composite art consisting of *Gīta* (melodic forms), *Vādya* (forms for drumming) and *Nṛtya* (dance, literally movements of the limbs of the body). *Saṅgīta* is of two kinds—*Mārga-saṅgīta* and *Deśī-saṅgīta*. The music for the mass was *Deśī Saṁgīta*. The construction and techniques of playing musical instruments like 14 kinds of drums and other rhythm-instruments have been vividly discussed.

In the medieval period, with the advent of Muslims, classical music in India began to evolve as two distinct traditions — (i) Hindustani Music and (ii) Carnatic Music. Hindustani Music spread to northern, eastern and western parts of India and Carnatic Music to the whole of Southern or Deccan Plateau region. There was a difference in the language, singing style, *svarsthānas,* rhythmic (*Tāla*) patterns and melodic structures owing to regional contexts and political changes. Both temple music and *Darbārī Saṁgī ta* came into being in the northern region. The classical tradition of Southern India maintained its purity and tradition restoring it in the sanctity of the temples. Many Indian and non-Indian cultures took an active part in this transformation. The advent of Islam at the end of the twelfth century brought Persian music and culture in the northern, eastern and western parts of the country. Personalities such as Amir Khusro, Raja Man Singh Tomar, Mian Tansen, Swami Haridas, Baiju Bavra, Gopal Nayak contributed in the development of Hindustani Classical Music during this period.

In the *Bhakti* movement, literature and music played a major role in propagating philosophies of human life. The works of composers like Jayadeva (eleventh century), Vidyapati (A.D. 1375), Chandidas (fourteenth to fifteenth century), Bhakta Narasimha (A.D. 1416–1475) and Meerabai (A.D. 1555–1603), Kabir and Tulsidas (fourteenth and fifteenth century), Surdas, Vallabhacharya and Chaitanya (seventeenth century) had a very strong impact on music traditions and practices during this period.



Musician: Mughal Painting of 17th Century
Source: CCRT

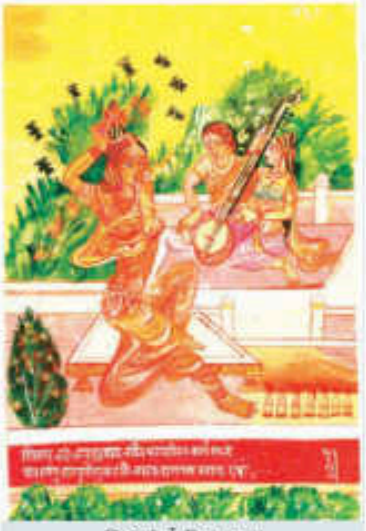








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Rāgiṇī Basant
Source: CCRT









Development of musical forms

In ancient period, *Jāti Gāyana* developed into *Prabandha gāyana*. Later in medieval period musical forms like, *Dhrupada, Dhamāra, Khayāl, Tarānā,* etc., evolved from *Prabandha*. Music which was accompanying drama had evolved now as an autonomous art form. New styles in instrumental music such as *Masītkhānī* and *Rāzkhanī* were developed in the medieval period. Pt. Somnath in his text *Rāga Vivodha*, described two fold description of *Rāga*, i.e., *Devamāyā svarūpa* (describing ethos of the *Rāga*) and *Nādamaya svarūpa*—tonal structure of the *Rāga*. *Rāga's* poetic description for better understanding of the ethos of a *Rāga*, lead to the creation of *Dhyāna Mantras*. These *dhyānas* were later depicted through *Rāga-mālā* paintings.

Rāgas were evolved from *Jāti lakṣaṇa*.
Jāti lakṣaṇa were adopted as *Rāga lakṣaṇa*.
Every *Rāga* has set number and sequences of *Svaras*.
In medieval period *Rāga dhyāna* tradition emerged.
Ragas are performed according to the time prescribed to them.
There are different varieties of classification of *Rāgas*.

Many new instruments like *Sitār, and Tablā* were developed. Amir Khusro, a very famous musician, is believed to have evolved many musical forms like *Qawwālī, Qual, Kalbana,* etc., *Rāgas* like *Yaman, Sāzgirī, Tālas* such as *Chapaka, Fartost,* etc. Under the patronage of kings, artists were encouraged to practice the intricacies of musical forms to refine their skills to higher levels of perfection. Gradually, musical forms started developing their traditions and styles by the efforts of practicing artists. This resulted into establishing *'Gharānā System'*. *Gharānā* is a term used to explain the belonging to a legacy of musical traditions either by lineage or by practicing a particular musical style. Every *Gharānā* has its own distinct features which emerge from mannerism and application. The concept of *Guru-Śiṣya* (Teacher-Pupil) leads to the sustainability of *Gharānā*.

Teaching in *Gharana* system is based upon oral tradition (*Guru-Śiṣya Paramparā*).
Gharana must have three generations to attain recognition.
There are many *Gharānās*. One of the important *Gharānās* is named after Mian Tansen—the famous court musician during Akbar period.

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Modern Period (A.D. 1800 – Till date)

In this period, Indian music flourished in the courts of kings. Few foreign scholars such as Sir William Jones, Sir W. Ouseley and Captain C.R. Day, Captain N.A. Willard showed a great aptitude for Hindustani Music and they wrote many valuable books on music. During this period Muhammad Raza (1813) wrote an important treatise *Naghmat-e-Asafi*. He is believed to be the first to adopt the *Bilawal Scale* as is *Shuddha Scale*.

Pt. V.N. Bhatkhande and Pt. V.D. Paluskar worked hard for the upliftment of music and musicians by organising music conferences, establishing music institutions and writing books. Efforts were made by Pt. V.N. Bhatkhande and Pt. V.D. Paluskar for documentation of music by evolving systematised Notation System.

This period is known for a revolutionary change in field of music introducing it in formal education system. Many scholars of this period have worked hard to get recognised as a discipline parallel to other disciplines.

Famous musicians of Modern period are Balkrishna Bua Ichalkaranjikar, Ustad Faiyaz Khan, Sawai Gandharva, Inayat Khan, Barkatulla Khan, Mushtaq Ali Khan, Nissar Hussain Khan, Allaudin Khan, Bade Gulam Ali Khan, Krishna Rao Shankar Pandit, Acharya Brihaspati, Omkar Nath Thakur and Vinayak Rao Patvardhan.

Pt. V.D. Paluskar started Gandharva Mahavidyalaya in Lahore (at present Pakistan).

Pt. V.N. Bhatkhande started Marris Music College in Lucknow, Uttar Pradesh, India.

Pt. V. N. Bhatkhande introduced *Thaat Raga Paddhati*.

Theatre

As a child all of us have played ghar-ghar (playing house). This way we playfully observed, imitated, became any character of choice, and had great fun. This is often referred as dramatic instinct in human beings. Drama is derived from a Greek word 'drama' meaning 'to do', 'to act'. Aristotle defined it as 'imitated human life'. Drama is also often called 'play' (*khel* or *natak khelna* in Hindi). Adults also play drama albeit in different ways. All of us play multiple roles in day-to-day life: one single person playing the role of a mother at home, teacher at school, passenger in a bus, etc. This may be called role play in real life. In formal theatre, drama is enacted on stage and there are two groups who participate—(i) actors and (ii) audience.

Dramatic enactments have been part of communities from prehistoric times, and each civilisation had developed






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its own norms of theatre. "The origin of Indian drama, as of significant drama anywhere else in the world, probably lies in the tribal ritual dances and celebration of antiquity. What began as community participation would gradually break down into two groups—those who perform and those who watch, that is the actors and the audience".—Som Benegal[1]

In India, there developed a rich tradition in theatre, particularly the classical Sanskrit drama. *Natyaśāstra*[2], ascribed to Sage Bharata, the oldest treatise on dramaturgy, gives an interesting account of the origin of Indian *Nāṭya* tradition in the opening chapter.









Mythological Story on the Origin of Indian Theatre

It is said that the four *Vedas*—as repository of knowledge and wisdom were accessible to upper castes and classes (*Vaṃas*), and not to women and lower caste or class. Dissatisfied with this situation, the Gods went to Brahmā, the creator, for a solution. The creator suggested the creation of the Fifth *veda* (*Pañcamaveda*) as *Nāṭyaveda*. This task was accomplished by Sage Bharata by way of extracting the wisdom of *Rigveda*, performative rituals of *Sāmaveda*, musicality of *Yajurveda* and emotive representations of *Atharvaveda*. The voluminous work, thus created, embodied all physical, theoretical and conceptual ideas behind traditional Indian drama. *Nāṭyaśāstra*, thus, speaks about the thought process and behaviour of people, their temperament, problems in society, needs, sorrow, etc., paving the way and promise for a better life.[3] It is recommended to go through important texts written by Shaktibhadra, Neelakantha Kalidasa, Bhattanarayana, Viswanatha and Kaviraja, who incorporated elements of *Nāṭyaśāstra* in their original writing.

Nāṭyaśāstra was written around the A.D. second century. According to the text (Chapter 35), a theatre group should have persons specialised in seventeen types of works: *Bharata* (Manager or producer or a multidimensional person), *Vidūṣaka* (Jester), *Tauryka* (experts in playing musical instruments), *Naṭa* (actor-Dancer), *Sūtradhāra* (linking and interpreting the text), *Nāṭyakāra* (the playwright), *Nandī* (in praise of the Creator while invocating the play), *Nāyaka* (person in lead character), *Mukuṭakāra* (Mask makers), *Ābharaṇakāra* (person engaged in making ornaments for a performance), *Mālyakāra* (person engaged in making garlands/ornaments), *Veṣakāra* (costume makers), *Chitrakāra* (painter/artist), *Rajaka* (person engaged in cleaning the costumes), *Kārukara* (sculptor-decorator) and *Kuśīlava* (actors in role proficient as actor-dancer-musicians). This list helps us to understand the components of a theatre group. Interestingly Nāṭyaśāstra includes names of each and every allied work considered necessary for a performance. For that reason, the *Rajakas* or *Mālyakāras* were honoured as a member of a theatre group, although they were not directly related to performances.

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1. Think: While reading the above paragraph of specialised people connected with different types of work for theatre, what comes to your mind?
2. Imagine as if you are in the second century and discuss whether in your opinion we who live in the twenty first century—are we a developed society?

The early drama was a combination of mime, poetry, prose, dialogue, humour, songs and dance. This made theatre as an all inclusive art form. Gradually it became an important medium for entertainment, learning, communication and reflexive thinking on human conditions.

The tradition of theatre in India certainly developed much before *Nāṭyaśāstra*; but systematic thinking or theorising about theatre arts started taking shape with *Nāṭyaśāstra*. It is believed that *Nāṭyaśāstra* had developed by compiling and conceptualising the prevailing folk traditions. The dramatic representations were conceived in three distinct forms; *Nāṭya* or the verbal act inclusive of singing; *Nṛtya* the non-verbal mimetic language, and *Nṛtta* or the language of body or dancing. For Bharata, the composite term *Abhinaya* would mean—*Abhi,* meaning learning from the play and *ni,* meaning to carry it forward. It intends to entertain and educate people simultaneously. When we talk of arts intervention in education today, we must critically judge its implications in the present times.

Ancient drama tradition can be categorised in two ways—*Nataka* which is *lokadharmi* or popular, and *nāṭyadharmi* or conventional. Sanskrit plays had a *Sūtradhāra* who created a link of events in a play through commentary, musical interlude or interpretative extempore.

Nāṭyaśāstra was written in Sanskrit. Kalidasa, the most celebrated poet of the Gupta era, wrote plays in Sanskrit and followed the *Nāṭyaśāstra* tradition. This establishes the fact that *Nāṭyaśāstra* was a pre-Gupta text. Another important point to be noted is the aspect of dependence of Sanskrit theatre on royal patronage. Necessarily, the audience of this theatre was social elite.

Interestingly, in Sanskrit drama characters could speak differently in classified dialects. The hero or the principal male characters speaking in Sanskrit, the heroine and principal female character in Prākrt; the royal attendants, servants and traders, rogues and villains, intriguers, cowherds, the people of the forest, and so on, in their respective dialects. This reveals the social categorisation of people of that time[1].






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Ashvaghosha

One can count about 35 plays from the classical Sanskrit genre but there are many more mentioned by others that are yet to be discovered. All the available ones are not classics.

Ashvaghosha

Palm-leaf fragments found in Central Asia take us back to the Kushan period of the A.D. first century. The earliest Sanskrit play, an unfinished one, comes in the form of a nine-act Buddhist play by Ashvaghosha, the court poet of the Kushan King Kanishka. This earliest Sanskrit play is called *Sāriputra Prakaraṇa*. Kalidasa, the most recognised poet-playwright who lived sometime in the fifth century in the glorious Gupta period of Indian history penned such classics as *Mālavikāgnimitram*, and *Abhijñāna-Śākuntalam* which are performed even today. Shudraka wrote *Mṛcchakatikam*, which is undoubtedly a masterpiece. Vishakhadatta's *Mudrārākṣasam* was written two hundred years later in the A.D. ninth century. One must remember Bhavabhuti, a scholarly Brahmin, proficient in rhetoric and logic, has left three works: two of which may be called historical-mythological—*Mahāvira-caritam* and *Uttararāma-caritam*—both dealing with the Ramayana, and the third with a romance titled *Mālati-Mādhavam*. The Sanskrit plays of Bhasha, Kalidasa, Shudraka, Harsha and Bhavabhuti complete a representative cross-section of Sanskrit drama.

The glory of Sanskrit drama declined through ninth to twelfth centuries, in the face of political turmoil. The language was also a big factor of its losing popularity.



Sanskrit Play

The decline of Sanskrit drama did not mean the decline of drama of Indian theatre. The growth of the *Prakṛts* made Sanskrit rather irrelevant and these popular languages threw up their own literatures and with it came a trend of powerful folk theatre tradition which survived the varied time and periods of history. This trend of theatre survives right down to this day in many parts of the country.

We find a few references of Urdu theatre in the court of Wazid Ali Shah who was a great patron

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of all performing arts. Although the ancient literature of the South, notably Tamil literature, has many references to drama and dramatic performances, there is little evidence of written plays with authorship. A form known as *Kuttu* meaning play or entertainment is found in Tamil literature. The *Kuttu* persists to this day in many forms of folk. There is the *therukuttu* form which appears to be a wholly rural artistic tradition. The most splendid and enduring contribution of the South to Indian theatre is storytelling traditions through dance and dance-drama. Notable among these are: *Bharatanatyam*, *Kathakali*, *Bhāgavata Mela Nāṭakam*, *Yakṣagaṇa* and *Kuchipudi*. The *Bhāgavata Mela Nāṭakam*, *Yakṣagaṇa* and *Kuchipudi* are said to be the regional variations of Indian Classical Theatrical tradition and they are sturdily dramatic in form employing special dazzling elaborate costumes and make-up. Although the *Bhāgavata Mela Nāṭakam* originally came from Tamil Nadu, its greatest manifestation is found in Andhra Pradesh which is also the home of *Kuchipudi*. On the contrary, *Kathakali* is a product of Kerala and perhaps the most vigorous and dramatic of all Indian dance-drama forms. The essence of all these dramatic forms is in the intersection of drama, dance, mime, gesture and music. In that sense, it may be considered as excellent examples of theatre *in toto*.



Sanskrit Play









Modern Developments

With the coming of the colonial rule, we find a strong influence of western, particularly English theatre tradition in the making of modern Indian theatre. It had interesting features: though essentially urban, it incorporated many classical and folk features and it simultaneously helped in the rise of commercial theatre and played a crucial role in raising nationalist sentiments. One may mention the names of Girish Chandra Ghosh and Bharatendu Harishchandra

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in this context. What followed was a good mixture of entertainment, nationalism and social reform.

One interesting development was Bombay Parsi theatre. Its initiator was a shrewd and resourceful man named Pestonji Framji who started the 'Original Theatrical Company' around 1870. His success led to the establishment of as many as six other flourishing companies, some of them breakaways from earlier groups. Although Bombay was their base, these companies travelled widely over North Indian and one of them, the Victoria Theatrical Company of Khurshidji Balliwala, even ventured as far as Britain.

We find simultaneous development of Bengali, Marathi, Kannad and other regional theatres together with the revival of many classical forms particularly in dance drama theatre. From 1930s to post independence period saw the rise of innovative forays theatre, music and dance. Uday Shankar's ballet, Rukmini Devi Arundale's Kalashetra attracted international attention and the rise of Indian People's Theatre Association (IPTA) changed the face of modern theatre forever.

Drawing upon the classical, folk and world theatre practices, IPTA paved the way for experimentations in developing new form and content in tackling emerging realities of post independent India. IPTA had inspired an enormous body of talents who subsequently enriched all the art fields including dance, music, literature, theatre and even cinema.

Mention may be made of some post independent personalities who shaped the contemporary Indian theatre: Sombhu Mitra, Chandrashekhar Kambar, Utpal Dutt, Habib Tanvir, Badal Sarkar, Mohan Rakesh, K.N. Panikkar, Vijay Tendulkar, Ibrahim Alkazi, Ratan Thyam, Girish Karnad, to name only a few. To understand the very recent developments in Indian theatre, one needs to study and understand the specific contributions of these people. We also need to look into the way Saṅgīta Nāṭaka Akademi and National School of Drama — two Government institutions, have played their roles in shaping contemporary Indian theatre. In recent years, theatre is playing interesting role in connecting education and social action.



Yakshagana performance

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We also need to appreciate the role played by folk theatre in shaping our Indian sensitivities—*Bhavai* from Gujarat, *Tamasha* from Maharashtra, *Bhand Pather* of Kashmir, *Jātrā* of Bengal, *Nautanki* of North India, *Yakṣagaṇa* of Karnataka, to name only a few. These are adding to the meaning of Indian theatre.

Dance in India

Dance is perhaps the oldest of all the living art forms of the human race. In fact cave paintings of prehistoric times display stylised movements or dance, conveying daily activities such as hunting, food gathering, enjoyment, etc. Images excavated from the remains of ancient civilisations give us valuable information about the history of our dance, such as the figurine of a bronze dancing girl which has been found in the Indus Valley civilisation relics.



Dancing Girl figure made of bronze
Source: https://nroer.gov.in









Dance in India has been an integral part of every social and cultural event. In a country such as ours, in which people of different races, languages and culture reside, there is a wide variety of forms in dance. Dances performed in India are believed to have their origins in dances and rituals of the period of the *Vedas*, which date back to around 5000 B.C.E. Bhimbetka Caves in the Vindhya Hill region of Madhya Pradesh are filled with images made by hunter-gatherers of dancers in various poses and group choreographies. Rather, as with most other branches of knowledge, its origin is ascribed to divinity. *Naṭarāja* or *Śiva* in the image of the Lord of Dance is revered as the supreme dancer. Some of the ancient treatise on dance and drama is the *Nāṭyaśāstra*, *Abhinaya Darpana*, *Nartana Sarvaswam*, *Rāmāyaṇa*, *Mahābhārata* and *Harivaṁśa*. Recorded evidences of dance have been found from the study of Anthropology, Archaeology, Sculpture, Paintings, Inscriptions, Religious and Secular Treatises and Literary Works.

Paintings of Ancient and Medieval India are also a storehouse of knowledge for the dance historian.

Gestures, body postures, expressions, etc., are the basic elements of Indian dance. The use of the hands and fingers (hand gestures called *mudrās*), eyes, head movement, formation or creation of different forms and shapes to the body are of primary importance in any dance form. The costume is of extreme importance where the clothes, jewellery and make-up showcase the identity of a particular place or region of India, e.g., in Bihu dance we see the men and women wearing typical Assamese dresses like *mekhola chaddar*, dhoti, whereas in the state of Rajasthan we find dancers wearing *lehenga* and *chunri*.

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The *Navrasas*— All forms have traditionally been structured around the nine *rasas* (emotions), which are *Śṛngāra* (love or beauty), *Hāsya* (laughter), *Karuṇā* (sorrow), *Raudra* (anger), *Vgīra* (heroism or courage), *Bhayānaka* (terror or fear), *Bibhatsa* (disgust), *Adbhuta* (surprise or wonder), and *Śānta* (peace or tranquility).

The essential elements of Indian dance are:

- the technique of movement, which includes facial expressions, head movements and body movements.
- all types of vocal and musical instruments and literature from Hindu mythological and ancient texts linked with dance.
- costumes, make-up and sets.

There are certain features that are common to all these dance styles. They all incorporate the three elements of Nartana or dance which are *Nṛtta* **(pure dance) or abstract,** Nṛtya **or expressive and** Nāṭya or dramatic dance. Being a communicative art forms, the role of abhinaya is very important which means 'to carry across ideas to the audience'. Sentiments, feelings and even moral precepts are carried across. Abhinaya is of four kinds:

Āṅgikābhinaya comprises of movements of the body or aṅga.

Vāchikābhinaya consists of *vachana* or words, songs, speeches and narration.

Āhāryābhinaya includes costumes, jewellery, make up, stage props and setting, that which has been acquired or aharita.

Sātvikābhinaya includes inner spiritual feelings, emotions and psychic states—expressed in a visible outward form.

1. What are the basic elements of dance?
2. View two different types of photograph of dancers or click two photos. Note the uniqueness in each.
3. Did you go to a historical monument and see sculptures of dance? Write a note in about 150 words.
4. Learn the *Navrasas* and recite them. Reflect whether these *rasas* fabricate your life.

The oldest evidence of Indian dance are descriptions of dance found in Barhut, Sanchi and Amaravati, which dates back to the second century B.C.E. Sculptures from all over India illustrates the importance of dance in Indian history and the richness of its traditions. It is said that Gautam Buddha, while being a king, had dancers, singers

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and instrument players in his court. In the case of Ajanta and Ellora, the paintings depict various mudrās of dances. During the classical Gupta age from the A.D. fourth to sixth century many sculptures give evidence of dance. The plain outer walls were decorated with narrative panels as well as dancing divinities. This was the beginning of a development that was to lead to the flourishing of dance images in Hindu temple architecture. During the so called 'medieval' period, approximately from the seventh to the sixteenth century, the most abundant representations of dance images can be seen in the Hindu temples of South India, in the Bhubaneshwar temples in East India, and in the temples of Khajuraho in central India. The West Indian Jain temples of Mt. Abu are also famous for their dance imagery.

One of the famous carvings of ninth century gateways is the Shiva temple in Chidambaram. What makes these Chidambaram *karana* reliefs so particular is that they are accompanied by the inscriptions of Sanskrit verses from the *Nāṭyaśastra* (a Sanskrit text on performing arts by Bharat Muni estimated to be written between five hundred B.C.E. and five hundred C.E.). Thus they form a kind of an illustrated dance manual carved in stone.






Many classical forms of dances are based on ancient sculptures. India has a very rich tradition of classical, tribal regional and folk dance forms. Our community life vibrates with singing and dancing. The birth of a child, weddings, festivals, the harvest, different seasons, or just the leisure that a field-worker enjoys at the end of the day, are occasions that call for dance as a celebration.

Living dance traditions

Tribal Dance

The dance of primitive and tribal groups generally performed by nomads who are close to nature fall in this category. Primitive ritualistic dance forms such as those of the Santhals, the Theyyam of Kerala in which masks of tigers and lions are used, the dance of the *Kalbelia* people who are the snake charmers of Rajasthan, the fire dance of Rajasthan and Gujarat, etc.

There are also certain Martial Art Dance forms like Kalaripayattu from Kerala, Gatka from Punjab, Mardani khel from Maharashtra and the like.

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Folk dances

These originated when humans started cultivating crops, and hence had to stay in fixed area and settlements. The dances centered around the agricultural calendar, especially the harvest and also around social events such as marriages, child birth and death. Dances of this category are the *Bihu* of Assam, the *Dandiya Raas* and *Garba* of Gujarat, the *Bhangra* of Punjab, the *Chari* of Rajasthan, etc.

The tribal and folk dance forms are basically participative dance forms in which everyone takes part and there is no distinction between the artist and the audience.









Classical Dances

Classical dances are sophisticated dance form polished by the intellect of accomplished artists and teachers who belong to intellectual or knowledgeable society.

Modern Dance

There are new styles of dance being evolved by accomplished dancers who are trained in the forms and aesthetics of the classical styles. This has also been termed as Creative, Innovative and Modern Dance.

Classical or Traditional Dance Forms of India

It is believed that these dances originated around the time when Janapadas were founded in India. They convey refined techniques of dance, story or message. Each classical dance form varies according to regional differences, social conditions, language, music, costumes, and of course, the individual contribution of artistes and teachers. The major classical or traditional dances of India, as accepted today, are:

1. Bharatanātyam
2. Kathak
3. Kathakali
4. Manipuri
5. Odissi
6. Kuchipudi
7. Mohiniattam
8. Sattriya

Kathak

A dance form with its roots in the northern part of India, it derives its name from the word *Katha*, or story. The journey from the rustic, wandering minstrel, to the storyteller of

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the temples of North India, through the courts of emperors and kings to the modern proscenium stage—the journey of Kathak is long and diversified. Lucknow Gharana, Jaipur Gharana, Banaras Gharana and Raigarh Gharana are the main gharanas or styles of Kathak.



Kathak Dance

Bharatanāṭyam

This dance form originated in southern part of India in the state of Tamil Nadu. It started as a temple dance tradition called Dasiyattam (the dance of the maid-servants) 2000 years ago. The name Bharatanâtyam is a simple derivation from the four most important aspects of dance (in Sanskrit). These are: *Bha* from *Bhâva* meaning emotion, *Ra* from *Râga* meaning music or melody, *Ta* from *Tăla* meaning rhythm and *Nāṭyam* meaning dance.



Bharatanatyam

Source: https://nroer.gov.in

Odissi

This dance form takes its name from the state of its origin, Odisha. It is approximately 60 years ago that Odissi, took its present form as seen today. Today's Odissi has evolved from the endless sculptures in various motifs carved on the temple walls of Orissa. Tribhangi, a unique posture in the form, literally meaning three bends in the body is inspired by temple sculptures creating an illusion of sculpture coming to life.

Kathakali

This is a dance form of Kerala which dates to around the seventeenth century and has become the most recognised icon of Kerala. The word Kathakali literally means 'story-play'. Kathakali is known for its heavy, elaborate make-up and costumes. This dance form presents themes derived from the Hindu epics, mythologies and legends. Initially Kathakali was thought to be the sole domain of men but now women also perform the dance.

Mohiniattam

Mohiniattam, the female dance form of kerala with its origin in the temples of Kerala was shaped to its present form by the Travancore Maharaja Kartika Thirunal and Swati Thirunal around nineteenth century. The dance form which was lost, was revived in through the dedicated efforts of Vallatol in twentieth century. *Lāsya* (languid grace) and Śṃgāra (beauty and love) are the essence and the basic mood of Mohiniattam. In Mohiniattam, it is compulsory to wear white costumes.



Odissi

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Kuchipudi

1. Imagine you are a storyteller and express an incident around you non-verbally.
2. *Do you know any Bharatanatyam dancer? Interact with them or interview them and give a report.*
3. *Name five* Odissi dancers, *and make a presentation on them.*
4. Find out about the costume and elaborate make-up of Kathakali.

Kuchipudi

This is a dance style adapted from the dance dramas enacted by Brahmin priests or *Bhagavutulus* whose dance was again influenced by the style of the Rājanartakīs or women court dancers of the time. The dance style is named after the place of its origin—Kuchipudi, a quiet village, about 65 kilometers from Vijayawada in Andhra Pradesh.

Manipuri

The classical dance of Manipur—the Jeweled City, is associated with Pre-Vaishnavite and Vaishnavite tradition. This dance form is a unique blend of Saṅkīrtana and Rāsalīlā. The dance with the drum called *Pung,* is one of the most vibrant and exciting presentations in Manipuri dance.



Sattriya

Sattriya

Sattriya or *Sattriya Nṛtya*, originated in the eastern state of Assam. It is a dance-drama performance art with origins in the Krishna centred Vaishnavism monasteries of Assam, and attributed to the fifteenth century Bhakti movement scholar and saint Mahapurush Srimanta Sankardev. One-act plays of *Sattriya* are called *Aṅkīya Nat*, which combine the aesthetic and the religious through a ballad, dance and drama. The plays are usually performed in the dance community halls (*namghar*) of monastery temples (*sattras*). The themes played relate to Krishna and Radha, sometimes other Vishnu *avatāras* such as Rama and Sita.

New styles of dance are being evolved by accomplished dancers who are trained in the forms and aesthetics of the classical styles. These have variously been termed as Creative, Innovative and Modern Dance. Creative genius like Uday Shankar, Chitralekha, Narendra Sharma, etc., evolved their own dance style taking the best from classical and folk dances and blending mythological and modern themes to suit the tastes of the modern audience.

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It is Rabindranath Tagore's efforts that made dance in India a socially acceptable art form. Tagore actively promoted dance and composed songs and dance dramas which he helped to choreograph and stage. He encouraged his students to learn various forms of dance and portray them with involvement and grace.

Some Folk forms (Performing Arts) of India

Have you heard of these Folk Dances of India—*Kalbelia*, *Cheraw* dance, Lama dance of Sikkim, and *Rouf* dance? Find out their States of origin and the important dancers who perform these dances.

Kalbelia: Folk form of Rajasthan

Kalbelia is a community of snake charmers living in Thar Desert Region of India. *Kalbelias* are now found predominantly in the districts of Jodhpur, Jaisalmer, Jalore and Barmer in western Rajasthan and in the cities of Jaipur and Pushkar in eastern Rajasthan. The women in flowing skirts dance to the beat of the *khanjari*, a percussion instrument, and the *poongi*, a wind instrument.

Cheraw

This dance form performed in Mizoram is recognised as one of the oldest forms of folk arts. It is characterised by the use of bamboo staves which are kept in cross and horizontal forms on the ground. While the male dancers move these bamboo staves in rhythmic beats, the female dancers perform by stepping in and out of the bamboo blocks.

Lama Music and Dances of Sikkim

Lama dances are masked dances performed with beats of varied types of rhythm by the Buddhist monks of Sikkim as a part of their religious practices. The practices are codified in the religious texts of Sikkim in accordance with the teachings of Guru Padmashambhava, the legendary saint of the Mahayana school of Buddhism. Based on these, Lamas conduct prayers for the benefit of Buddhism and the state of Sikkim. Gorgeous attire and colourful masks made according to the scriptures are used with traditional musical instruments like cymbals and big horns rendering the codified religious music and chanting.

Rouf

One of the most popular traditional music and dances of Kashmir is *Rouf*. It graces all the festive occasions, especially






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Eid and *Ramzan* days. The music and dance is performed by a group of women facing each other. The song that is sung during the performance is in the form of question and answer where one group questions and the other responds to it in a rhythmic way.

Shadow Puppet Theatre Traditions of India

The communities which are distinctively associated with it are:

1. *Chamadyacha Bahulya Thakar,* Maharashtra
2. *Tolu Bommalatta Killekyata or Are Kapu,* Andhra Pradesh
3. *Togalu Gombeyatta Killekyata or Dayat,* Karnataka
4. *Tolu Bommalattam Killekyata,* Tamil Nadu
5. *Tolpava Kuthu Vellalachetti, Nair,* Kerala
6. *Ravanachhaya Bhat,* Orissa

Though these forms have distinct regional identities, languages and dialects in which they are performed, they share a common worldview, aesthetics and themes. The narratives are mainly based on the epics of *Rāmāyaṇa* and *Mahābhārata*, *Purāṇas*, local myths and tales. They communicate significant messages to the rural community besides entertainment.

Some examples of folk or living traditions of theatre

Bhavai

Bhavai is a traditional theatre form of Gujarat and Rajasthan with its deep roots in Kutch and Kathiawar in Gujarat. Bhavai originated as a prayer to the Goddess *Amba*, during the Navaratra festival.

Tamasha

Tamasha is a traditional folk theatre form of Maharashtra. The term 'Tamasha' is derived from Persian.

Bhagawatmela and Kuchipudi

Many dance drama forms in Andhra Pradesh and Tamil Nadu are prevalent which are called by different names—*Bhāgavatamela*, *Kuchipudi*, *Bhamakalapam*, *Yakṣagaṇa*, etc. Others, such as *Vithināṭakam* of Andhra Pradesh and *Terukoothu* in Tamil Nadu are street theatre forms. Besides, there are several *Kurvanji* forms.

Bhand Pather

Bhand theatre is popular in Northern India, specially in Kashmir and Punjab. *Bhand Pather* of Kashmir, is a unique combination of dance, music and drama where satire, wit and parody are preferred for inducing laughter.

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Nautanki

As the term *Nautanki* itself suggests, the basis of this form of entertainment is *Natak* (play). *Nautanki* presentation of Kanpur, Lucknow, Hathras and a few places in Haryana are famous because of their distinctive styles. Parsi theatre also influenced this traditional theatre form. The most important role of narrator is played by the *Ranga*, who begins the story to be presented. The *Ranga* connects the various episodes of the play. The most important instruments of *Nautanki* are the *Nagaara* and *Tikaari*.



Swang

Swang is a folk-theatre form of Haryana and Punjab. It is similar to *Nautanki*, traditional theatre form of Uttar Pradesh. *Bahurupias* also practised *Swang* to earn their livelihood. On the occasion of Holi festival, *Swang* artists move around in small groups displaying their art and during Dussehra festival, they are part of various tableaux.

Rāslīlā

Based exclusively on the legendary stories of Radha and Krishna, *Rāslīlā* is a highly musical traditional theatre form of Uttar Pradesh.









Jātrā

Jatra is one of the most popular traditional theatre forms of Bengal and Orissa. They travel from one place to another to perform hence the name *Jātrā*. Over the centuries, faith and religious fervour has been replaced in this folk theatre form with contemporary themes of social and historical significance.

Kutiyattam

In Kerala, the tradition of *Kutiyattam* is easily the most prominent survivor among the forms containing some essential elements of content and structural features of Sanskrit theatre. While scholars are correct in often calling *Kutiyattam* the only surviving tradition in Sanskrit theatre, *Kutiyattam* has also incorporated local traditions, culture and themes.



Bhāgavatamela and Kuchipudi

Many dance drama forms in Andhra Pradesh and Tamil Nadu are prevalent which are called by different names—*Bhagawatmela*, *Kuchipudi*, *Bhamakalapam*, Yakshagana, etc. Others, such as *Veethinatakam* of Andhra Pradesh and *Terukoothu* in Tamil Nadu are street theatre forms.

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